Coordination Compounds

(2026)

1. The secondary valency of Co in the complex $[Co(NH_3)_5(NO_2)]^{2+}$ is

(A) 5 **(1 Marks) (CBSE 2026-56/1/1)**

(B) 1

(C) 4

(D) 6

2. (a) Write IUPAC names of the following coordination compounds :

(i) $[Ag(NH_3)_2][Ag(CN)_2]$ **(2 Marks) (CBSE 2026-56/1/1)**

(ii) $K_3[Fe(C_2O_4)_3]$

OR

(b) (i) Give a chemical test to show that $[Co(NH_3)_5SO_4]Cl$ and $[Co(NH_3)_5Cl]SO_4$ are ionisation isomers.

(ii) What is meant by the 'Chelate effect'? Give an example.

3. The Valence Bond Theory (VBT) explains the formation, magnetic behaviour and geometry of coordination compounds. The Crystal Field Theory (CFT) of coordination compounds is based on the effect of different crystal fields (provided by the ligands taken as point charges), on the degeneracy of d-orbital energies of the central metal atom/ion. The splitting of the d-orbitals provides different electronic arrangements in strong and weak crystal fields.

Answer the following questions : **(4 Marks) (CBSE 2026-56/1/1)**

(a) In octahedral crystal field, energies of which d-orbitals will be raised when ligands approach the central metal atom/ion ? Give reason in support of your answer.

(b) Using crystal field theory, write the electronic configuration of central metal atom/ion of the following :

(i) $[CoF_6]^{3-}$

(ii) $[Co(NH_3)_6]^{3+}$ [At. No. : Co = 27]

(c) $[NiCl4\ ]^{2-}$ is paramagnetic while $[Ni(CO)_4]$ is diamagnetic though both are tetrahedral. Why ? [Atomic No. : Ni = 28]

OR

(c) Write hybridization and magnetic behaviour of the complex

$[Fe\ (CN)_6]^{3-}$ [Atomic No. : Fe = 26]

Answer

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1. (D)

2. (a) (i) Diamminesilver(I) dicyanidoargentate(I)
(ii) Potassium trioxalatoferrate (III)

OR

(b) (i) $[Co(NH_3)_5(SO_4)]Cl$ with $AgNO_3$ solution gives white ppt of AgCl whereas $[Co(NH_3)_5Cl]SO_4$ does not / $[Co(NH_3)_5Cl]SO_4$ with $BaCl_2$ gives white ppt of $BaSO_4$ whereas $[Co(NH_3)_5(SO_4)]Cl$ does not.

(ii) Formation of stable complexes by di or polydentate ligand with single metal atom/ion.

For example: $[Co(en)_3]^{3+}$ (or any other correct example)

3. (a)

- dx^2-y^2 and dz^2
- In Octahedral Crystal field, ligands approach metal atom/ion along the axis due to which dx^2-y^2 and dz^2 orbitals experience more repulsion.

(b)

(i) $[CoF_6]^{3-}$ $Co^{3+} = 3d^6$ $t_{2g}^4\ eg^2$

(ii) $[Co(NH_3)_6]^{3+}$ $Co^{3+} = 3d^6$ $t_{2g}^6\ eg^o$

(c) In $[NiCl_4]^{2-}$, Cl^- is a weak field ligand and does not pair up the unpaired electrons in 3d-orbital and hence paramagnetic.
In $[Ni(CO)_4]$, CO being a strong field ligand pair up the unpaired electrons and hence diamagnetic.

OR

(c) Hybridisation- d^2sp^3
Magnetic behaviour: Paramagnetic

(2025)

Q.1 The correct IUPAC name of $[Pt(NH_3)_2Cl_2]^{2+}$ is

(1 Mark) (CBSE 2025 - 56/1/1)

A. Diamminedichloridoplatinum (O)

B. Diamminedichloridoplatinum (II)

C. Diamminedichloridoplatinate (IV)

D. Diamminedichloridoplatinum (IV)

Q.2 Assertion (A) : $[Cr(H_2O)_6]Cl$ and $[Fe(H_2O)_6]Cl2$ are examples of homoleptic complexes.
Reason (R) : All the ligands attached to the metal are the same.

(1 Mark) (CBSE 2025 - 56/1/1)

A. Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

B. Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

C. Assertion (A) is false, but Reason (R) is true.

D. Assertion (A) is true, but Reason (R) is false.

Q.3 The number of moles of AgCl precipitated when excess $AgNO_3$ solution is mixed with one mole of $[Co(NH_3)_3Cl_3]$ is :

(1 Mark) (CBSE 2025 - 56/7/1)

A. 1

B. 2

C. 3

D. 0

Q.4 The complex ions $[Co(NH_3)_5(NO_2)]^{2+}$and $[Co(NH_3)_5(ONO)]^{2+}$ are called

(1 Mark) (CBSE 2025 - 56/2/1)

A. Geometrical isomers

B. Linkage isomers

C. Co-ordination isomers

D. Ionization isomers

Q.5 The diamagnetic species is :

[At. No. Co = 27, Fe = 26, Ni = 28]

(1 Mark) (CBSE 2025 - 56/2/1)

A $[Fe(CN)_6]^{3-}$

B $[CoF_6]^{3-}$

C $[Ni(CN)_4]^{2-}$

D $[NiCl_4]^{2-}$

Q.6 Polyhalogen compounds have wide application in industries and agriculture. DDT is also a very important polyhalogen compound. It is a :

(1 Mark) (CBSE 2025 - 56/4/1)

A. fertilizer

B. greenhouse gas

C. non-biodegradable insecticide

D. biodegradable insecticide

Q.7 According to Werner's theory of coordination compounds :

(1 Mark) (CBSE 2025 - 56/6/1)

A. Both primary and secondary valences are ionisable.

B. Secondary valences are ionisable.

C. Both primary and secondary valences are non-ionisable.

D. Primary valences are ionisable.

Q.8 Which of the following complex ion is not optically active?

(1 Mark) (CBSE 2025 - 56/6/1)

A. $[Co(en)_3]^{3+}$

B. $[Co(ox)_3]^{3-}$

C. trans- $[Co(en)_2Cl^2]^+$

D. cis- $[Co(en)_2Cl_2]^+$

Q.9 Write the IUPAC name of the complex $[Pt(en)_2Cl_2]^{2+}$. Draw the structure of geometrical isomer of this complex which is optically inactive.

(2 Mark) (CBSE 2025 - 56/7/1)

Q.10 (i) Write the formula of the following coordination compound :
Pentaamminecarbonatocobalt(III)chloride
(ii) Write the IUPAC name of the linkage isomer of the complex $[Co(NH_3)_5(NO_2)]Cl_2$.

(2 Mark) (CBSE 2025 - 56/7/1)

Q.11 Name the following coordination compounds according to IUPAC norms :

(2 Mark) (CBSE 2025 - 56/2/1)

(a) $\left[Co(NH_3)_4\,(H_2O)\,Cl\right]\,Cll_2$
(b) $[CrCl_2(en)_2]\,Cl$

Q.12 PtCl4·2KCl doesn't give precipitate of AgCl with $AgNO_3$ solution. Write the structural formula and IUPAC name of the complex.

(2 Mark) (CBSE 2025 - 56/4/1)

Q.13 Write IUPAC names of the following coordination compounds:

(i) $[CoCl_2(en)_2]SO_4$
(ii) $K_3[Fe(C_2O_4)_3]$

(2 Mark) (CBSE 2025 - 56/6/1)

Q.14 Differentiate between :
(i) Double salt and Complex compound
(ii) Didentate ligand and Ambidentate ligand

(2 Mark) (CBSE 2025 - 56/6/1)

Q.15 (a) What is meant by crystal field splitting energy? For a d^4 ion, write the configuration if (i) $\Delta_0 < P$, and (ii) $\Delta_0 > P$.

(3 Mark) (CBSE 2025 - 56/5/1)

(b) Explain why in tetrahedral coordination entities, low spin configurations are rarely observed.

Q.16 Using valence bond theory, explain the hybridization and magnetic character of the following :

(3 Mark) (CBSE 2025 - 56/2/1)

(a) $\left[Co(NH_3)_6\right]^{3+}$
(b) $[Ni(CO)_4]$
[At. no. : $Co = 27, Ni = 28$]

Q.17 Answer the following about the complexes $[FeF_6]^{3-}$ and $[Fe(CN)_6]^{4-}$:
(i) Write the hybridization involved in each case.
(ii) Which of them is the outer orbital complex and which one is the inner orbital complex ?
(iii) Compare their magnetic behaviour.
[Atomic number : Fe = 26]

(3 Mark) (CBSE 2025 - 56/6/1)

Q.18 (i) What happens to the colour of complex $[Ti(H_2O)_6]^{3+}$ when heated gradually?

(ii) Write the electronic configuration for d^5 ion if $\Delta_o < P$.
(iii) Write the hybridization and magnetic behaviour of the complex $[Ni(CO)_4]$.
[Atomic number : Ni = 28]

(3 Mark) (CBSE 2025 - 56/6/1)

Q.19 Werner's coordination theory in 1893 was the first attempt to explain the bonding in coordination complexes. It must be remembered that this theory was put forward before the electron had been discovered by J.J. Thomson in 1897, and before the electronic theory of valency. Werner did not have any of the modern instrumental techniques and all his studies were made using simple experimental techniques. Werner was able to explain the nature of bonding in complexes and he concluded that in complexes, the metal shows two different sorts of valency : primary and secondary. Primary valences are normally ionisable whereas secondary valences are non ionisable.

(4 Mark) (CBSE 2025 - 56/5/1)

Answer the following questions :

(a) One mole of $CrCl_3 \cdot 4H_2O$ precipitates one mole of AgCl when treated with excess of $AgNO_3$ solution. Write (i) the structural formula of the complex, and (ii) the secondary valency of Cr.

(b) What is the difference between a complex and a double salt ?

(c) (i) Arrange the following complexes in the increasing order of conductivity of their solution :

$$\left[Cr(NH_3)_3Cl_3\right], \quad \left[Cr(NH_3)_6\right]Cl_3, \quad \left[Cr(NH_3)_5Cl\right]Cl_2$$

Question 2
Werner's coordination theory in 1893 was the first attempt to explain the bonding in coordination complexes. It must be remembered that this theory was put forward before the electron had been discovered by J.J. Thomson in 1897, and before the electronic theory of valency. Werner did not have any of the modern instrumental techniques and all his studies were made using simple experimental techniques. Werner was able to explain the nature of bonding in complexes and he concluded that in complexes, the metal shows two different

sorts of valency : primary and secondary. Primary valences are normally ionisable whereas secondary valences are non ionisable.

Answer the following questions :

(c) (ii) Write two differences between primary and secondary valences in coordination compounds.

Q.20 The Crystal Field Theory (CFT) of coordination compounds is based on the effect of different crystal fields (provided by the ligands taken as point charges) on the degeneracy of d-orbital energies of the central metal atom/ion. The splitting of the d-orbitals provides different electronic arrangements in strong and weak crystal fields. In tetrahedral coordination entity formation, the d-orbital splitting is smaller as compared to the octahedral entity.

(4 Mark) (CBSE 2025 - 56/7/1)

Answer the following questions :

(a) On the basis of CFT, explain why $[Ti(H_2O)_6]Cl_3$ complex is coloured ?

What happens on heating the complex $[Ti(H_2O)_6]Cl_3$?

Give reason.

[Atomic no. : Ti = 22]

(b) (i) What is crystal field splitting energy ?

Question 2
(b) (ii) On the basis of Δ_0 and P (pairing energy), how can you differentiate between a strong field ligand and a weak field ligand ?

(c) Why are low spin tetrahedral complexes rarely observed ?

Q.21

Answer the following :

(a) Low spin tetrahedral complexes are not known.

(b) Co^{2+} is easily oxidised to Co^{3+} in the presence of a strong ligand [At. No. of $Co = 27$]

(c) What type of isomerism is shown by the complex $[Co(NH_3)_6]\,[Cr(CN)_6]$?

(d) Why a solution of $[Ni(H_2O)_6]^{2+}$ is green while a solution of $[Ni(CN)_4]^{2-}$ is colourless. (At. No. of $Ni = 28$)

(e) Write the IUPAC name of the following complex : $[Co(NH_3)_5\,(CO_3)]\,Cl$

(5 Mark) (CBSE 2025 - 56/1/1)

Q.22 (a) What is meant by 'Chelate effect' ? Give an example.

(b) Write the hybridization and magnetic behaviour of $[Fe(CN)_6]^{4-}$. (Atomic number : Fe = 26)

(c) If $PtCl_2 \cdot 2NH_3$ does not react with $AgNO_3$, what will be its formula?

(5 Mark) (CBSE 2025 - 56/1/1)

Q.23

(i) Give reasons :

(I) $[Ni(CO)_4]$ is diamagnetic whereas $[NiCl_4]^{2-}$ is paramagnetic. [Atomic number : $Ni = 28$]

(II) CO is a stronger complexing agent than NH_3.

(III) The trans isomer of complex $[Co(en)_2Cl_2]^+$ is optically inactive.

(ii) Using Crystal Field theory, write the number of unpaired electrons in octahedral complexes of Fe^{3+} in the presence of :

(I) Strong field ligand

(II) Weak field ligand

[Atomic number : $Fe = 26$]

(5 Mark) (CBSE 2025 - 56/4/1)

Q.24 (i) Name the type of isomerism exhibited by the following compounds. Also draw their corresponding isomers.

(5 Mark) (CBSE 2025 - 56/4/1)

(I) $[Co(NH_3)_6]\,[Cr(CN)_6]$

(II) $[Co(en)_3]^{3+}$

(III) $[Co(NH_3)_3(NO_2)_3]$

(ii) Differentiate between weak field and strong field ligands. How does the strength of the ligand influence the spin of the complex?

Answer

Q.1 D

Q.2 B

Q.3 D

Q.4 B

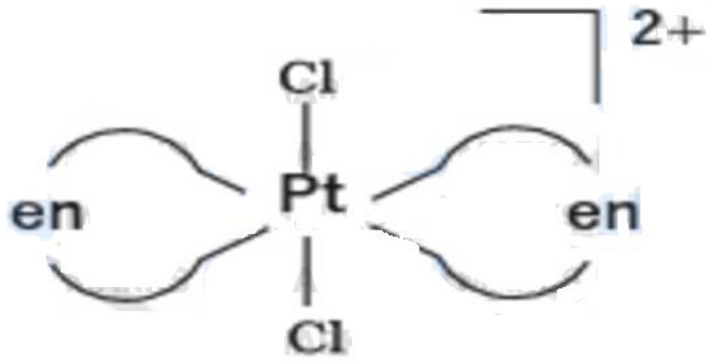
Q.5 C

Q.6 C

Q.7 D

Q.8 C

Q.9 a) Dichloridobis(ethane-1,2-diamine)platinum(IV) ion



Q.10 i) $[Co(NH_3)_5(CO_3)]Cl$

ii) Pentaamminenitrito-O-cobalt(III) chloride

Q.11 (a) Tetraammineaquachloridocobalt(III) chloride

(b) Dichloridobis(ethane-1,2-diamine)chromium(III) chloride

Q.12 Structural formula: $K_2[PtCl_6]$

IUPAC Name: Potassium hexachloridoplatinate(IV)

Q.13 (i) Dichloridobis(ethane-1,2-diamine)cobalt(IV) sulphate

(ii) Potassium trioxalatoferrate(III)

Q.14 (i) Double salts dissociate into simple ions while complex compounds do not dissociate completely into ions when dissolved in water. (Or any other suitable difference)

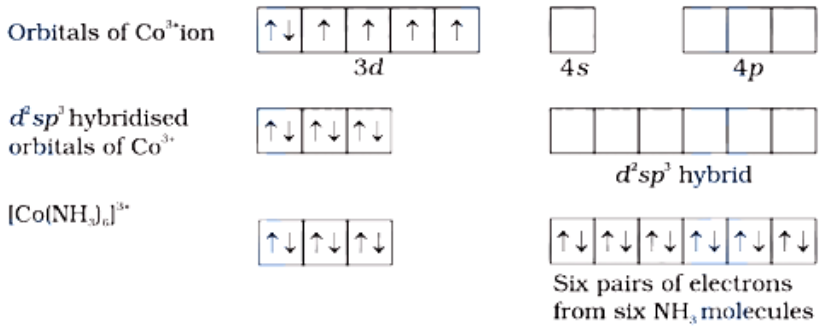
(ii) When a ligand binds through two donor atoms is called a didentate ligand while a unidentate ligand which has two different donor atoms and either of the two ligates in the complex is called ambidentate ligand.

Q.15 (a)The difference of energy between the two sets of d-orbitals t_2g and eg due to the presence of ligands in a definite geometry. / The energy required to split the degenerate d-orbitals into two sets of orbitals.

(i) $\mathbf{t^3_{2g}e^1_g}$
(ii) $\mathbf{t^4_{2g}e^0_g}$

(b) Orbital splitting energy is not sufficiently large for causing pairing of electrons.

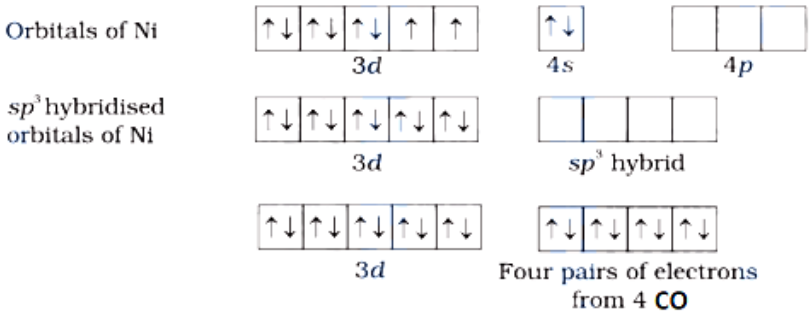
Q.16 (a) (b)(i)



Hybridization: d^2s^3

Magnetic character: Diamagnetic.

(b)



Hybridization: sp^3

Magnetic character: Diamagnetic.

Q.17 a)

$[FeF_6]^{3-}$ – $sp^3\ d^2$

$[Fe(CN)_6]^{4-}$ – d^2sp^3

b)

$[FeF_6]^{3-}$

-outer orbital complex

$[Fe(CN)_6]^{4-}$

- inner orbital complex

c)

$[FeF_6]^{3-}$

- paramagnetic

$[Fe(CN)_6]^{4-}$

-diamagnetic

Q.18

a) It becomes colourless/ colour slowly fades away

b) $t_{2g}^{3} e_g^{2}$

c) sp^3, diamagnetic

Q.19 (a)

(i) $[Cr(H_2O)_4Cl_2]Cl$

(ii) 6

(b) Double salts dissociate into simple ions while complex compounds do not dissociate completely into ions when dissolved in water. (Or any other suitable difference)

(c) (i) $\left[Cr(NH_3)_3Cl_3\right] < \left[Cr(NH_3)_5Cl\right]Cl_2 < \left[Cr(NH_3)_6\right]Cl_3$

Qus.2 (c)(ii)

Primary Valency	**Secondary Valency**
1. Ionisable	1. Non-ionisable
2. Satisfied by negative ions	2. Satisfied by negative ions or neutral molecules

(or any other two suitable differences)

Q.20 (a) Due to presence of one unpaired electron in t_{2g} which gets excited to e_g / Due to excitation energy $t_{2g}^{1} \rightarrow e_g^{1}$, it gives colour. (d-d transition)

When heated, water is lost therefore crystal field splitting does not occur and it becomes colourless.

(b) The energy required to split the degenerate d-orbitals into two sets of orbitals (t_{2g} and e_g. /The difference of energy between the two sets of d - orbitals t_{2g} and eg due to the presence of ligands in a definite geometry.

Qus 2. (b) (ii) $\Delta_0 < P$, weak filed ligand

$\Delta_0 > P$, strong field ligand

(c) Because the orbital splitting energies are not sufficiently large for forcing pairing / Due to low crystal field splitting energy.

Q.21 a) The orbital splitting energies are not sufficiently large for forcing paring of electrons.

b) In the presence of strong field ligand, d^7 is converted into more stable d^6 configuration / Strong field effect stabilises higher oxidation state.

c) Co-ordination isomerism.

d) $[Ni(H_2O)_6]^{2+}$ has unpaired electrons whereas $[Ni(CN)4]^{2-}$ has no unpaired electron.

e) Pentaamminecarbonatocobalt(III) chloride

Q.22 (a) The higher stability of complexes involving chelating ligands as compare to complexes having non-chelating ligand.

Example: $[Co(en)_3]^{3+}$ (or any other correct example)
(b) d^2sp^3, diamagnetic
(c) $[Pt(NH_3)_2Cl_2]$

Q.23 (a)(i)

(I)CO being a strong field ligand, causes pairing of electrons therefore, there is no unpaired electron.
Whereas $Cl^-$is a weak field ligand, does not cause pairing, therefore presence of unpaired electrons.

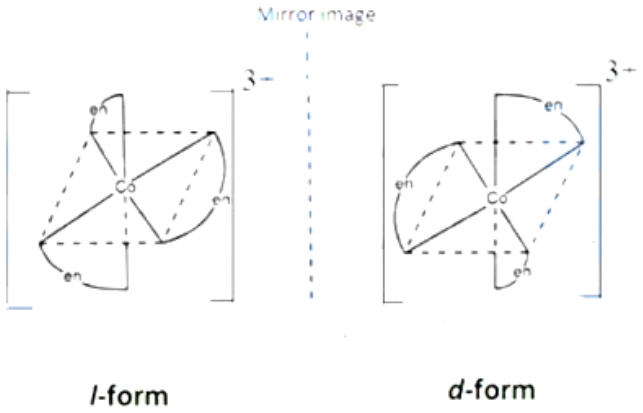
(II) CO can form both sigma (σ) and pi (π)bond with central metal atom/Metal to ligand bonding creates synergic effect between CO and the Metal.

(III) Mirror images are superimposable/ Presence of plane of symmetry.
(ii)
(I) $\Delta 0 > P$, causes pairing of electrons, therefore 1 unpaired electron
(II) $\Delta 0$

Q.24

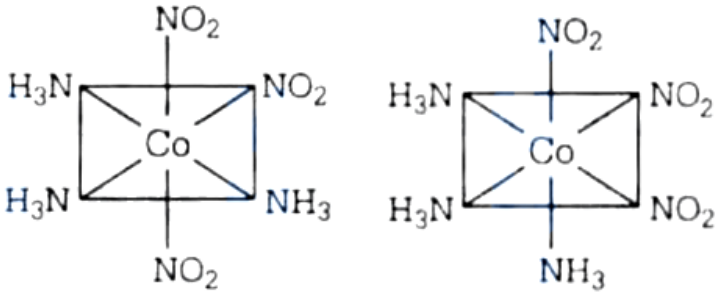
(I)Coordination Isomerism / $\left[Cr(NH_3)_6\right]\left[Co(CN)_6\right]$

(II)Optical Isomerism /



l-form *d*-form

(III)Geometrical isomerism /



(ii) Weak field ligands produce weak field and leads to small splitting of d-orbitals whereas strong field ligands produce strong field leading to large splitting of d-orbitals.

Strong field ligands cause pairing of electrons/a smaller number of unpaired electrons hence produces low spin complexes and weak field ligands causes no pairing of electrons/ a greater number of unpaired electrons hence produces high spin complexes.

(2024)

Q.1 The type of isomerism shown by the complex $[CoCl_2(en)_2]^+$ is :

(1 Mark) (CBSE 2024 - 56/3/1)

A. Linkage isomerism

B. Geometrical isomerism

C. Ionisation isomerism

D. Coordination isomerism

Q.2 Which of the following is diamagnetic in nature?

(1 Mark) (CBSE 2024 - 56/3/1)

A. Co^{3+}, in a square planar complex

B. Co^{3+}, in a tetrahedral complex
[Atomic number : Co = 27]

C. Co^{3+}, octahedral complex with strong field ligand

D. Co^{3+}, octahedral complex with weak field ligand

Q.3 Ligand $EDTA^{4-}$ is an example of a :

(1 Mark) (CBSE 2024 - 56/2/1)

A. Tridentate ligand

B. Monodentate ligand

C. Didentate ligand

D. Polydentate ligand

Q.4 Which of the following ligand forms chelate complex ?

(1 Mark) (CBSE 2024 - 56/2/1)

A $C_2O_4^{2-}$

B NH_3

C NO_2^-

D Cl^-

Q.5 Write the IUPAC names of the following coordination compounds (any three) :

(3 Mark) (CBSE 2024 - 56/5/1)

(a) $\left[Co(NH_3)_4Cl\,(NO_2)\right]\,Cl$
(b) $\left[Ni(NH_3)_6\right]\,Cl_2$
(c) $K_3\,\left[Cr(C_2O_4)_3\right]$
(d) $[Co(en)_2Br_2]^+$

Q.6 (a) Write the formula for the following coordination compound :
Potassium tetrahydroxidozincate (II)

(3 Mark) (CBSE 2024 - 56/1/1)

(b) Arrange the following complexes in the increasing order of conductivity of their solution :

$[Cr(NH_3)_5Cl]Cl_2$, $[Cr(NH_3)_3Cl_3]$, $[Cr(NH_3)_6]Cl_3$

(c) Identify the type of isomerism exhibited by the following complexes:
(i) $[Co(NH_3)_5NO_2]^{2+}$
(ii) $[Co(en)_3]Cl_3$

Q.7 (a) Draw the geometrical isomers of the given complex :
$[Pt(en)_2Cl_2]^{2+}$

(3 Mark) (CBSE 2024 - 56/4/1)

(b) Write the electronic configuration for d^4 ion if $\Delta_0 < P$ on the basis of crystal field theory.

(c) What is meant by a unidentate ligand ? Give an example.

Q.8 The Valence Bond Theory (VBT) explains the formation, magnetic behaviour and geometrical shapes of coordination compounds whereas 'The Crystal Field Theory' for coordination compounds is based on the effect of different crystal fields (provided by ligands taken as point charges), on the degeneracy of d-orbital energies of the central metal atom/ion. The splitting of the d-orbitals provides different electronic arrangements in strong and weak crystal fields. The crystal field theory attributes the colour of the coordination compounds to d − d transition of the electron. Coordination compounds find extensive applications in metallurgical processes, analytical and medicinal chemistry.

(4 Mark) (CBSE 2024 - 56/1/1)

(a) What is crystal field splitting energy ?

(b) Give reason for the violet colour of the complex $\left[\mathrm{Ti}(\mathrm{H_2O})_6\right]^{3+}$ on the basis of crystal field theory.

(c) $\left[\mathrm{Cr}(\mathrm{NH_3})_6\right]^{3+}$ is paramagnetic while $\left[\mathrm{Ni}(\mathrm{CN})_4\right]^{2-}$ is diamagnetic. Explain why. [Atomic No. : $\mathrm{Cr} = 24, \mathrm{Ni} = 28$]

Question 2

(c) Explain why $\left[\mathrm{Fe}(\mathrm{CN})_6\right]^{3-}$ is an inner orbital complex, whereas $\left[\mathrm{Fe}(\mathrm{H_2O})_6\right]^{3+}$ is an outer orbital complex.

[Atomic No.: $\mathrm{Fe} = 26$]

Q.9 The nature of bonding, structure of the coordination compound can be explained to some extent by valence bond theory. The central metal atom/ion makes available a number of vacant orbitals equal to its coordination number. The appropriate atomic orbitals (s, p and d) of the metal hybridise to give a set of equivalent orbitals of definite geometry such as square planar, tetrahedral, octahedral and so on. A strong covalent bond is formed only when the orbitals overlap to the maximum extent. The d-orbitals involved in the hybridisation may be either inner d-orbitals i.e. (n − 1) d or outer d-orbitals i.e. nd. The complexes formed are called inner orbital complex (low spin complex) and outer orbital complex (high spin complex) respectively. Further, the complexes can be paramagnetic or diamagnetic in nature. The drawbacks of this theory are

that this involves number of assumptions and also does not explain the colour of the complex.

(4 Mark) (CBSE 2024 - 56/4/1)

(a) Predict whether $[CoF_6]^{3-}$ is diamagnetic or paramagnetic and why ? [Atomic number : $Co = 27$]

(b) What is the coordination number of Co in $[Co(en)_2Cl_2]^+$?

(c) (i) Write the IUPAC name of the given complex :

$[Pt(NH_3)_2Cl_2]^{2+}$

(ii) Explain $[Co(NH_3)_6]^{3+}$ is an inner orbital or outer orbital complex.

Question 2

(c) Using valence bond theory, deduce the shape and hybridisation of $[Ni(NH_3)_6]^{2+}$ [Atomic number of $Ni = 28$]

Q.10 The oxidation number of the central atom in a complex is defined as the charge it would carry if all the ligands are removed along with the electron pairs that are shared with the central atom. Similarly the charge on the complex is the sum of the charges of the constituent parts i.e. the sum of the charges on the central metal ion and its surrounding ligands. Based on this, the complex is called neutral if the sum of the charges of the constituents is equal to zero. However, for an anion or cationic complex, the sum of the charges of the constituents is equal to the charge on the coordination sphere.

(4 Mark) (CBSE 2024 - 56/5/1)

(a) Define ambidentate ligand with an example.
(b) What type of isomerism is shown by $[Co(NH_3)_5Cl]SO_4$ and $[Co(NH_3)_5SO_4]$?
(c) Define Chelate effect. How it affects the stability of complex?

Qus 2. (c) Find the coordination number and oxidation state of chromium in $Na_3[Cr(C_2O_4)_3]$.

Q.11 Attempt any five of the following :

(5 Mark) (CBSE 2024 - 56/3/1)

(a) Write the IUPAC name of the complex :
$[Co(H_2O)(CN)(en)_2]^{2+}$

(b) Why is geometrical isomerism not possible in tetrahedral complexes having two different types of unidentate ligands coordinated with the central metal ion?

(c) Arrange the following complex ions in increasing order of their crystal field splitting energy (Δ_0) :

$\left[Co(NH_3)_6\right]^{3+}, [CoF_6]^{3-}, [Co(CN)_6]^{3-}$

(d) Write the hybridization and magnetic character of the complex $[Ni(CO)_4]$ on the basis of valence bond theory.

[Atomic No. : $Ni = 28$]

(e) Out of $[CoF_6]^{3-}$ and $\left[Co(C_2O_4)_3\right]^{3-}$, which one complex is :

(i) more stable ?

(ii) the high spin complex ?

(f) What is the difference between an ambidentate ligand and bidentate ligand?

(g) Write the electronic configuration of d^5 in terms of t_{2g} and e_g in an octahedral field when :

(i) $\Delta_o > P$, and

(ii) $\Delta_0 < P$

Q.12

Answer any five of the following :

(a) How is the crystal field splitting energy for octahedral complex (Δ_O) related to that of tetrahedral complex (Δ_t) ?

(b) Write the IUPAC name of the following complex :

$[PtCl_2(en)_2]\,(NO_3)_2$

(c) Write the geometry and magnetic behaviour of the complex $[Ni(CO)_4]$ on the basis of Valency Bond Theory (VBT).

(d) What type of isomerism is shown by the complex $\left[Co(NH_3)_6\right]\,[Cr(CN)_6]$?

(e) For the coordination compound on the basis of crystal field theory, write the electronic

configuration for d^4 ion if $\Delta_o < P$. Is the coordination compound a high spin or low spin complex ?

(f) Out of $[Co(NH_3)_6]^{3+}$ and $[Co(NH_3)_4Cl_2]^+$, which complex is heteroleptic and why ?

(g) Draw the structures of optical isomers of $[PtCl_2(en)_2]^{2+}$.

(5 Mark) (CBSE 2024 - 56/2/1)

Answer

Q.1 B

Q.2 C

Q.3 D

Q.4 A

Q.5 a) Tetraamminechloridonitrito-N-cobalt (III) chloride
b) Hexaamminenickel(II) chloride
c) Potassium trioxalatochromate (III)
d) Dibromidobis(ethane-1,2-diamine)cobalt (III) ion

Q.6

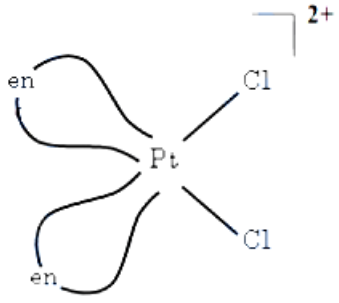
a) $K_2[Zn(OH)_4]$

(b) $[Cr(NH_3)_3Cl_3] < [Cr(NH_3)_5Cl]Cl_2 < [Cr(NH_3)_6]Cl_3$

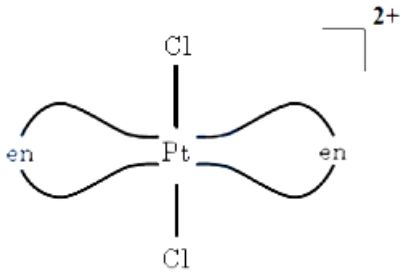
(c) (i) Linkage isomerism

(ii) Optical isomerism

Q.7 (a)



Cis isomer



Trans isomer

(b) $t_{2g}^3 e_g^1$

(c) When a ligand is bound to a metal atom or ion through a single donor atom.

Example: $Cl^-/H_2O/NH_3$ (or any other one correct example)

Q.8 (a) The energy used in the splitting of degenerate d - orbitals due to the presence of ligands in a definite geometry is called Crystal Field Splitting Energy.

(b) $Ti^{3+} = 3\,d^1$ i.e. $t_{2g}{}^1 e_g^0$ Due to $d - d$ transition.

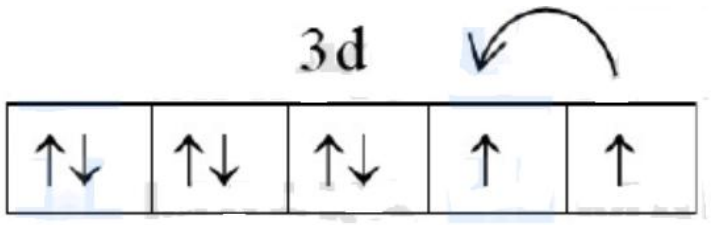
(c)

$Cr^{3+} = 3\,d^3$



Due to stable $t_{2g}{}^3$ configuration, hence paramagnetic.

$Ni^{2+} = 3\,d^8$



CN^- being strong field ligand pair up the electrons and hence diamagnetic.

Question 2

Solution: (c) CN^- being a strong ligand leads to the pairing of electrons in $[Fe(CN)_6]^{3-}$ leading to d^2sp^3 hybridization. H_2O being a weak ligand does not lead to the pairing of electrons in $[Fe(H_2O)_6]^{3+}$ leading to $sp^3\,d^2$ hybridization. / In $[Fe(CN)_6]^{3-}$, $(n-1)d$ orbitals of central metal ion are used in hybridization (d^2sp^3). Hence inner orbital complex whereas in $[Fe(H_2O)_6]^{3+}$ nd orbitals of central metal ion are used in hybridization $(sp^3\,d^2)$.

Q.9 (a) Paramagnetic, F^- does not cause pairing of electrons and hence unpaired electrons are left.

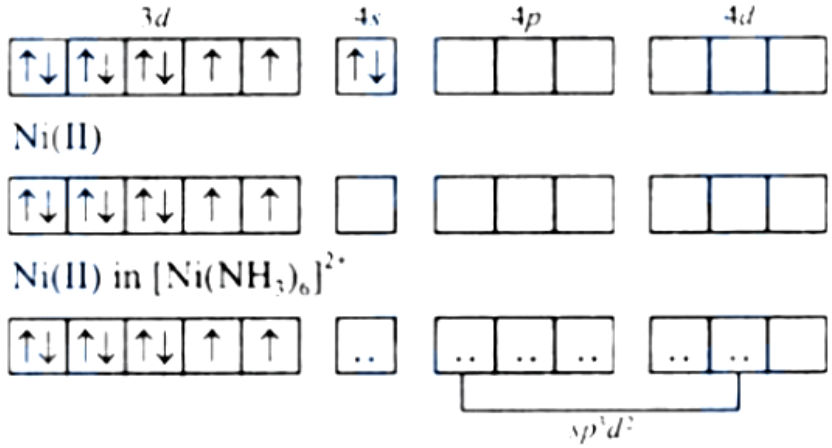
(b) 6

(c) (i) diamminedichloridoplatinum (IV) ion

(ii) It uses inner d orbitals because NH_3 causes pairing of electrons

Question 2

Solution: c)



Shape: Octahedral ; Hybridization : sp^3d^2

Q.10 a) Ligand which has two different donor atoms and either of the two ligates in the complex.

Examples :- $NO_2^-/SCN^-/CN^-$ (any one)

b) Ionization isomerism

c) When a bi-dentate or polydentate ligand ligates to the metal atom or ion that forms a ring. More is the chelation more is the stability.

Qus 2. c)Coordination number: 6 Oxidation state : +3

Q.11 (a) Aquacyanidobis(ethane-1,2-diamine)cobalt(III) ion

(b) Because of same relative positions of the unidentate ligands attached to the central metal atom.

(c) $[CoF_6]^{3-} < [Co(NH_3)_6]^{3+} < [Co(CN)_6]^{3-}$

(d) sp^3, diamagnetic

(e) (i) $[Co(C_2O_4)_3]^{3-}$

(ii) $[CoF_6]^{3-}$

(f) An ambidentate ligand is one which can ligate through either of the two different donor atoms while bidentate ligand is bound to metal through two donor atoms.

(g) (i) $t_{2g}^5 \quad e_g^0$ (ii) $t_{2g}^3 \quad e_g^2$

Q.12

(a) $\Delta_t = \left(\frac{4}{9}\right) \Delta_o$

(b) Dichloridobis(ethane -1,2-diamine)platinum (IV) nitrate

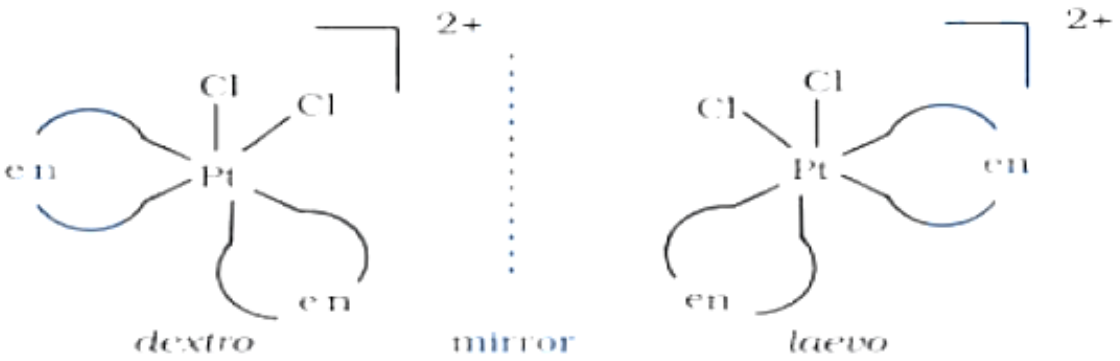
(c) sp^3, diamagnetic

(d) Coordination isomerism

(e) $t_{2g}^3 e_g^1$, high spin

(f) $[Co(NH_3)_4Cl_2]^+$, as the metal is bound to more than one donor group(ligand).

(g)



Previous Years' CBSE Board Questions

5.1 Werner's Theory of Coordination Compounds

MCQ

1. **Assertion :** $[Co(NH_3)_5SO_4]Cl$ gives a white precipitate with silver nitrate solution.
Reason : The complex dissociates to give Cl^- and SO_4^{2-} ions.
(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is true, but Reason (R) is false.
(d) Assertion (A) is false, but Reason (R) is true. *(2023)*

2. One mole of $CrCl_3 \cdot 6H_2O$ compound reacts with excess $AgNO_3$ solution to yield two moles of $AgCl_{(s)}$. The structural formula of the compound is
(a) $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$ (b) $[Cr(H_2O)_3Cl_3] \cdot 3H_2O$
(c) $[Cr(H_2O)_4Cl_2]Cl \cdot 2H_2O$ (d) $[Cr(H_2O)_6Cl_3]$ *(2020)*

VSA (1 mark)

3. When a coordination compound $CrCl_3 \cdot 6H_2O$ is mixed with $AgNO_3$, 2 moles of AgCl are precipitated per mole of the compound. Write structural formula of the complex. *(AI 2019, 1/2, Delhi 2016)*

4. What is the difference between a complex and a double salt? *(AI 2019)* R

5. When a coordination compound $CoCl_3 \cdot 6NH_3$ is mixed with $AgNO_3$, 3 moles of AgCl are precipitated per mole of the compound. Write structural formula of the complex. *(1/2, AI 2016)*

5.2 Definition of Some Important Terms Pertaining to Coordination Compounds

MCQ

6. The oxidation number of Co in $[Co(en)_3]_2(SO_4)_3$ is
(a) +2 (b) +3 (c) +4 (d) +6
(2021C) Ap

7. Which of the following ligands form a 'chelate' complex with metal ion ?
(a) H_2O (b) CN^- (c) $C_2O_4^{2-}$ (d) Cl^-
(2021C)

8. The coordination number of Cr in $[CrCl_2(ox)_2]^{3-}$ is
(a) 6 (b) 5 (c) 4 (d) 3
(2021C)

9. The coordination number of 'Co' in the complex $[Co(en)_3]^{3+}$ is
(a) 3 (b) 6 (c) 4 (d) 5
(2020) R

10. **Assertion (A) :** $[Pt(en)_2Cl_2]^{2+}$ complex is less stable than $[Pt(NH_3)_4Cl_2]^{2+}$ complex.
Reason (R) : $[Pt(en)_2Cl_2]^{2+}$ complex shows chelate effect.
(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement. *(2020)*

11. The oxidation state of Ni in $[Ni(CO)_4]$ is
(a) 0 (b) 2
(c) 3 (d) 4 *(2020)* An

VSA (1 mark)

12. Amongst $[Fe(C_2O_4)_3]^{3-}$ and $[Fe(NH_3)_6]^{3+}$, which is more stable and why? *(2019C)*

13. Write the coordination number and oxidation state of platinum in the complex $[Pt(en)_2Cl_2]$. *(2018)*

14. Which of the following is more stable complex and why?
$[Co(NH_3)_6]^{3+}$ and $[Co(en)_3]^{3+}$ *(Delhi 2014)*

5.3 Nomenclature of Coordination Compounds

MCQ

15. The formula of the coordination compound tetraamminechloridonitrito-N-cobalt(III) chloride is :
(a) $[Co(NH_3)_4Cl(ONO)]Cl_2$
(b) $[Co(NH_3)_4Cl_2(NO_2)]Cl$
(c) $[Co(NH_3)_4Cl(NO_2)]Cl$
(d) $[Co(NH_3)_4(NO_2)]Cl_3$ *(2021C)*

VSA (1 mark)

16. Write the formula of the following coordination compound :
Iron(III) hexacyanidoferrate(II) *(1/3, 2018)*

17. Write the IUPAC name of the following complex : $[Cr(NH_3)_3Cl_3]$. *(1/3, Delhi 2017)*

18. Write the IUPAC name of the following complex: $[Co(NH_3)_5(CO_3)]Cl$. *(1/3, Delhi 2017)*

19. When a coordination compound $CrCl_3 \cdot 6H_2O$ is mixed with $AgNO_3$, 2 moles of AgCl are precipitated per mole of the compound. Write IUPAC name of the complex. *(1/2, Delhi 2016)* An

SA I (2 marks)

20. Give the formulae of the following compounds :
    (a) Potassium tetrahydroxidozincate(II)
    (b) Hexaammineplatinum(IV) chloride *(2020)* Ap

21. Using IUPAC norms write the formulae for the following :
    (i) Pentaamminenitrito-O-cobalt(III) chloride
    (ii) Potassium tetracyanidonickelate(II)
    (NCERT, Delhi 2019)

22. Using IUPAC norms write the formulae for the following :
    (a) Sodium dicyanidoaurate(I)
    (b) Tetraamminechloridonitrito-N-platinum(IV) sulphate *(AI 2017)*

23. Using IUPAC norms write the formulae for the following :
    (a) *Tris*(ethane-1,2,diamine)chromium(III) chloride
    (b) Potassium tetrahydroxozincate(II) *(AI 2017)*

24. Using IUPAC norms write the formulae for the following :
    (a) Potassium trioxalatoaluminate(III)
    (b) Dichlorido*bis*(ethane-1,2-diamine)cobalt(III)
    (AI 2017) Ap

25. (i) Write down the IUPAC name of the following complex :
    $[Cr(NH_3)_2Cl_2(en)]Cl$ (*en* = ethylenediamine)
    (ii) Write the formula for the following complex :
    Pentaamminenitrito-O-cobalt (III). *(Delhi 2015)*

26. Using IUPAC norms write the formulae for the following coordination compounds :
    (i) Hexaamminecobalt(III) chloride
    (ii) Potassium tetrachloridonickelate(II) *(AI 2015)*

27. (i) Write down the IUPAC name of the following complex : $[Cr(en)_3]Cl_3$.
    (ii) Write the formula for the following complex : Potassium trioxalatochromate(III).
    (Foreign 2015)

SA II (3 marks)

28. (i) Write the IUPAC name of the following complex : $[Co(NH_3)_4(H_2O)Cl]Cl_2$
    (ii) What is the difference between an Ambidentate ligand and a Bidentate ligand ?
    (iii) Out of $[Fe(NH_3)_6]^{3+}$ and $[Fe(C_2O_4)_3]^{3-}$, which complex is more stable and why?
    (Term II, 2021-22) U

29. Write the IUPAC name of the following :
    (i) $[Co(NH_3)_6]Cl_3$ (ii) $[NiCl_4]^{2-}$
    (iii) $K_3[Fe(CN)_6]$ *(AI 2015C)* Ap

30. Write down the IUPAC name for each of the following complexes :
    (i) $[Co(NH_3)_5Cl]Cl_2$ (ii) $K_3[Fe(CN)_6]$
    (iii) $[NiCl_4]^{2-}$ *(Delhi 2014C)* U

31. Write the IUPAC name and draw the structure of each of the following complex entities :
    (i) $\left[Co\begin{pmatrix}COO\\|\\COO\end{pmatrix}_3\right]^{3-}$ (ii) $[Cr(CO)_6]$
    (iii) $[PtCl_3(C_2H_4)]^-$
    (At. nos. Cr = 25, Co = 27, Pt = 78) *(AI 2014C)* Cr

5.4 Isomerism in Coordination Compounds

Structural Isomerism

MCQ

32. The compounds $[Co(SO_4)(NH_3)_5]Br$ and $[Co(Br)(NH_3)_5]SO_4$ represent
    (a) optical isomerism
    (b) linkage isomerism
    (c) ionisation isomerism
    (d) coordination isomerism. *(2023)*

33. The pair $[Co(NH_3)_4Cl_2]Br_2$ and $[Co(NH_3)_4Br_2]Cl_2$ will show
    (a) linkage isomerism
    (b) hydrate isomerism
    (c) ionization isomerism
    (d) coordinate isomerism *(2020)* U

34. **Assertion (A)** : Linkage isomerism arises in coordination compounds because of ambidentate ligand.
    Reason (R) : Ambidentate ligand like NO_2 has two different donor atoms *i.e.*, N and O.
    (a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
    (b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
    (c) Assertion (A) is correct, but Reason (R) is incorrect statement.
    (d) Assertion (A) is incorrect, but Reason (R) is correct statement. *(2020)*

VSA (1 mark)

35. What type of isomerism is shown by the complex $[Co(NH_3)_5NO_2]Cl_2$? *(2020)*

36. What type of isomerism is exhibited by the complex $[Co(NH_3)_5Cl]SO_4$? *(1/3, 2018)*

    OR

    What type of isomerism is exhibited by the following complex : $[Co(NH_3)_5SO_4]Cl$ *(Foreign 2014)*

37. What type of isomerism is shown by the complex $[Co(NH_3)_6][Cr(CN)_6]$?
    (1/3, Delhi 2017, Foreign 2014)

38. What type of isomerism is shown by the complex $[Co(NH_3)_5(SCN)]^{2+}$? *(1/3, AI 2017)*

39. What type of isomerism is exhibited by the complex $[Co(NH_3)_5NO_2]^{2+}$? *(Foreign 2014)*

SA I (2 marks)

40. (a) Which of the following species cannot act as a ligand? Give reason.
OH^-, NH_4^+, CH_3NH_2, H_2O
(b) The complex $[Co(NH_3)_5(NO_2)]Cl_2$ is red in colour. Give IUPAC name of its linkage isomer.
(2023)

Stereoisomerism

VSA (1 mark)

41. What type of isomerism is shown by the complex $[Co(en)_2Cl_2]^+$? Name the structure of an isomer of this complex which is optically active. *(1/2, 2020)* An

42. What type of isomerism is shown by the complex $[Co(en)_3]Cl_3$? *(NCERT Intext, 1/3, Delhi 2017)*

OR

What type of isomerism is exhibited by the complex $[Co(en)_3]^{3+}$? (*en* = ethane-1,2-diamine) *(1/3, AI 2014)*

43. Draw one of the geometrical isomers of the complex $[Pt(en)_2Cl_2]^{2+}$ which is optically active.
(1/3, Delhi 2016)

44. Draw one of the geometrical isomers of the complex $[Pt(en)_2Cl_2]^{2+}$ which is optically inactive. *(AI 2016)*

45. Draw the geometrical isomers of complex $[Pt(NH_3)_2Cl_2]$. *(1/3, Delhi 2015)*

46. Write down the IUPAC name of the complex $[Pt(en)_2Cl_2]^{2+}$. What type of isomerism is shown by this complex? *(AI 2015)*

47. Draw the geometrical isomers of complex $[Pt(en)_2Cl_2]^{2+}$. *(1/3, Foreign 2015)* Cr

SA I (2 marks)

48. Write IUPAC name of the complex $[Pt(en)_2Cl_2]$. Draw structures of geometrical isomers for this complex. *(2019)*

49. Write IUPAC name of the complex $[Cr(NH_3)_4Cl_2]^+$. Draw structures of geometrical isomers for this complex. *(Delhi 2019)*

SA II (3 marks)

50. Indicate the types of isomerism exhibited by the following complexes :
(i) $[Co(NH_3)_5(NO_2)]^{2+}$
(ii) $[Co(en)_3]Cl_3$ (*en* = ethylene diamine)
(iii) $[Pt(NH_3)_2Cl_2]$ *(Delhi 2015C)* U

51. Write the IUPAC name of the complex $[Cr(NH_3)_4Cl_2]^+$. What type of isomerism does it exhibit? *(Delhi 2014)*

52. Draw the structures of optical isomers of each of the following complex ions :
$[Cr(C_2O_4)]^{3-}$, $[PtCl_2(en)_2]^{2+}$, $[Cr(NH_3)_2Cl_2(en)]^+$
(Delhi 2014C)

5.5 Bonding in Coordination Compounds

Valence Bond Theory

MCQ

53. The magnetic moment of $[NiCl_4]^{2-}$ is
(a) 1.82 B.M. (b) 2.82 B.M.
(c) 4.42 B.M. (d) 5.46 B.M.
[Atomic number : Ni = 28] *(2023)*

VSA (1 mark)

54. Write the IUPAC name and hybridisation of the complex, $[CoF_6]^{3-}$.
(Given : Atomic number of Co = 27) *(1/2, 2020)*

OR

Write the hybridisation and number of unpaired electrons in the complex $[CoF_6]^{3-}$. (Atomic no. of Co = 27) *(1/3, 2018)*

55. Predict the number of unpaired electrons in the square planar $[Pt(CN)_4]^{2-}$ ion. *(2019C)* An

56. Write the hybridisation and magnetic character of $[Co(C_2O_4)_3]^{3-}$.
(At. no. of Co = 27) *(1/3, Delhi 2017)*

57. Why is $[NiCl_4]^{2-}$ paramagnetic but $[Ni(CO)_4]$ is diamagnetic? (At. no. : Cr = 24, Co = 27, Ni = 28)
(NCERT Intext, 1/3, AI 2014)

SA I (2 marks)

58. (i) Write the IUPAC name of $[Mn(H_2O)_6]SO_4$.
(ii) Why is $[Fe(CN)_6]^{4-}$ diamagnetic while $[FeF_6]^{3-}$ is paramagnetic ? [At. No. Fe= 26] *(2021C)*

59. Write IUPAC name and hybridization of the following complexes :
(i) $[Ni(CO)_4]$
(ii) $[CoF_6]^{3-}$
(Atomic number Ni = 28, Co = 27) *(2020)* Ap

60. (a) Write the IUPAC name and hybridisation of the complex $[Fe(CN)_6]^{3-}$.
(Given : Atomic number of Fe = 26)
(b) What is the difference between an ambidentate ligand and a chelating ligand ? *(2020)* Ap

61. Out of $[CoF_6]^{3-}$ and $[Co(en)_3]^{3+}$, which one complex is
(i) paramagnetic (ii) more stable
(iii) inner orbital complex and
(iv) high spin complex
(Atomic no. of Co = 27) *(2019)*

62. Out of $[CoF_6]^{3-}$ and $[Co(C_2O_4)_3]^{3-}$, which one complex is
(i) diamagnetic
(ii) more stable
(iii) outer orbital complex and
(iv) low spin complex ?
(Atomic no. of Co = 27) *(Delhi 2019)*

63. For the complex $[Fe(H_2O)_6]^{3+}$, write the hybridization, magnetic character and spin of the complex. (At. number Fe = 26) *(2/3,AI 2016)*

64. Write the hybridization and shape of the following complexes :
(i) $[CoF_6]^{3-}$
(ii) $[Ni(CN)_4]^{2-}$
(Atomic number : Co = 27, Ni = 28) *(AI 2015)*

65. Write the state of hybridization, shape and IUPAC name of the complex $[CoF_6]^{3-}$. (Atomic no. of Co = 27) *(Foreign 2014)*

66. Write the state of hybridization, shape and IUPAC name of the complex $[Ni(CN)_4]^{2-}$. (Atomic no. of Ni = 28) *(Foreign 2014)*

67. Write the state of hybridization, shape and IUPAC name of the complex $[Co(NH_3)_6]^{3+}$. (Atomic no. of Co = 27) *(Foreign 2014)*

SA II (3 marks)

68. (a) On the basis of crystal field theory, write the electronic configuration for d^4 with a strong field ligand for which $\Delta_o > P$.
(b) A solution of $[Ni(H_2O)_6]^{2+}$ is green but a solution of $[Ni(CO)_4]$ is colourless. Explain.
[Atomic number : Ni = 28] *(2023)*

69. (a) Write the hybridisation, shape and magnetic character of $[CoF_6]^{3-}$.
[Atomic number of Co = 27]
(b) Write the formula of pentaamminechloridocobalt(III) chloride. *(Term II, 2021-22)* Ev

70. (i) Calculate the spin only magnetic moment of the complex $[FeF_6]^{3-}$. (Atomic number of Fe = 26)
(ii) Write the IUPAC name of the given complex : $[Co(NH_3)_5Cl]Cl_2$
(iii) Why is the complex $[Co(en)_3]^{3+}$ more stable than $[CoF_6]^{3-}$? *(Term II, 2021-22)* Ev

71. (a) Using valence bond theory, predict the hybridization and magnetic character of the complex : $[Ni(CO)_4]$ (Atomic number : Ni = 28)
(b) Write IUPAC name of $[Pt(NH_3)_2Cl(NO_2)]$.
(c) Why $[Co(en)_3]^{3+}$ is a more stable complex than $[Co(NH_3)_6]^{3+}$? *(Term II, 2021-22)* Ev

72. Write the hybridisation and magnetic character of the following complexes :
(i) $[Fe(H_2O)_6]^{2+}$
(ii) $[Ni(CN)_4]^{2-}$
[Atomic number : Fe = 26, Ni = 28] *(AI 2019)*

LA (5 marks)

The following questions are case-based questions. Read the case carefully and answer the questions that follow.

73. In coordination compounds, metals show two types of linkages, primary and secondary. Primary valencies are ionisable and are satisfied by negatively charged ions. Secondary valencies are non-ionisable and are satisfied by neutral or negative ions having lone pair of electrons. Primary valencies are non-directional while secondary valencies decide the shape of the complexes.
(i) If $PtCl_2.2NH_3$ does not react with $AgNO_3$, what will be its formula?
(ii) What is the secondary valency of $[Co(en)_3]^{3+}$?
(iii) (1) Write the formula of Iron(III)hexacyanidoferrate(II)
(2) Write the IUPAC name of $[Co(NH_3)_5Cl]Cl_2$.

OR

(iii) Write the hybridization and magnetic behaviour of $[Ni(CN)_4]^{2-}$. *(2023)*

Crystal Field Theory

MCQ

74. **Assertion (A) :** Low spin tetrahedral complexes are rarely observed.
Reason (R) : Crystal field splitting energy is less than pairing energy for tetrahedral complexes.
(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is wrong statement.
(d) Assertion (A) is wrong, but Reason (R) is correct statement. *(2023, 2020)* R

75. The crystal field splitting energy for octahedral (Δ_o) and tetrahedral (Δ_t) complexes is related as
(a) $\Delta_t = \frac{2}{9}\Delta_o$ (b) $\Delta_t = \frac{5}{9}\Delta_o$
(c) $\Delta_t = \frac{4}{9}\Delta_o$ (d) $\Delta_t = 2\Delta_o$ *(2020)*

VSA (1 mark)

76. Why a solution of $[Ni(H_2O)_6]^{2+}$ is green while a solution of $[Ni(CN)_4]^{2-}$ is colourless?
(At. no. of Ni = 28) *(NCERT, 1/3, Delhi 2017)* Ev

SA I (2 marks)

77. (a) Why is $[NiCl_4]^{2-}$ paramagnetic while $[Ni(CN)_4]^{2-}$ is diamagnetic?
(Atomic number of Ni = 28)
(b) Why are low spin tetrahedral complexes rarely observed? *(2/3, AI 2017)* An

78. (i) On the basis of crystal field theory, write the electronic configuration of d^4 ion if $\Delta_o < P$.
(ii) Write the hybridization and magnetic behaviour of the complex $[Ni(CO)_4]$.
(At. no. of Ni = 28) *(2/3, Delhi 2015)* An

SA II (3 marks)

79. (i) On the basis of crystal field theory, write the electronic configuration for d^4 ion if $\Delta_o < P$.
(ii) Using valence bond theory, predict the hybridization and magnetic character of $[Ni(CN)_4]^{2-}$. (Atomic number of Ni = 28)

(iii) Write the formula of the following complex using IUPAC norms : Dichloridobis (ethane-1,2-diamine) cobalt (III) *(Term II, 2021-22)*

80. (i) Write the electronic configuration of d^4 on the basis of crystal field splitting theory, if $\Delta_o < P$.
(ii) $[Ni(CN)_4]^{2-}$ with square-planar structure is diamagnetic and $[NiCl_4]^{2-}$ with tetrahedral geometry is paramagnetic. Give reason to support the statement.
[Atomic number : Ni = 28]
(iii) Write the number of ions produced in the solution from the following complex :
$[PtCl_2(NH_3)_4]Cl_2$ *(Term II, 2021-22)* Ap

81. (a) Write the IUPAC name of the following complex : $K_3[Cr(C_2O_4)_3]$
(b) On the basis of crystal field theory, write the electronic configuration of d^5 ion if $\Delta_o < P$.
(c) What are ambidentate ligands?
(Term II, 2021-22)

82. (i) Write the IUPAC name of the following complex : $K_2[PdCl_4]$
(ii) Using crystal field theory, write the electronic configuration of d^5 ion, if $\Delta_o > P$.
(iii) What are homoleptic complexes ?
(Term II, 2021-22)

83. (i) Why chelate complexes are more stable than complexes with unidentate ligands?
(ii) What is "spectrochemical series"? What is the difference between a weak field ligand and a strong field ligand ? *(Term II, 2021-22)*

84. (a) Give the IUPAC name and electronic configuration of central metal atom in terms of t_{2g} and e_g of $K_4[Mn(CN)_6]$.
(b) What is meant by 'Chelate effect' ? Give an example. *(2020)* U

85. Define the following :
(a) Ambidentate ligands
(b) Spectrochemical series
(c) Heteroleptic complexes *(2019C)*

5.6 Bonding in Metal Carbonyls

VSA (1 mark)

86. Out of NH_3 and CO, which ligand forms a more stable complex with a transition metal and why?
(1/3, AI 2015) U

5.7 Importance and Applications of Coordination Compounds

MCQ

87. **Assertion (A) :** EDTA is used to determine hardness of water.
Reason (R) : EDTA is a bidentate ligand.
(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is true, but Reason (R) is false.
(d) Assertion (A) is false, but Reason (R) is true.
(2023)

CBSE Sample Questions

5.1 Werner's Theory of Coordination Compounds

MCQ

1. The number of ions formed on dissolving one molecule of $FeSO_4.(NH_4)_2SO_4.6H_2O$ in water is
(a) 3 (b) 4
(c) 5 (d) 6 *(2022-23)*

5.2 Definition of Some Important Terms Pertaining to Coordination Compounds

MCQ

2. Ambidentate ligands like NO_2^- and SCN^- are
(a) unidentate
(b) didentate
(c) polydentate
(d) has variable denticity. *(2020-21)* R

5.3 Nomenclature of Coordination Compounds

MCQ

3. The formula of the coordination compound tetraammineaquachloridocobalt(III) chloride is
(a) $[Co(NH_3)_4(H_2O)Cl]Cl_2$
(b) $[Co(NH_3)_4(H_2O)Cl]Cl_3$
(c) $[Co(NH_3)_2(H_2O)Cl]Cl_2$
(d) $[Co(NH_3)_4(H_2O)Cl]Cl$ *(2020-21)*

5.4 Isomerism in Coordination Compounds

SA I (2 marks)

4. The formula $Co(NH_3)_5CO_3Cl$ could represent a carbonate or a chloride. Write the structures and names of possible isomers. *(2022-23)*

5.5 Bonding in Coordination Compounds

MCQ

5. The CFSE of $[CoCl_6]^{3-}$ is 18000 cm^{-1}, the CFSE for $[CoCl_4]^-$ will be
   (a) 18000 cm^{-1} (b) 8000 cm^{-1}
   (c) 2000 cm^{-1} (d) 16000 cm^{-1}
   (2022-23)

SA I (2 marks)

6. (i) Using Crystal Field Theory, write the electronic configuration of iron ion in the following complex ion. Also predict its magnetic behaviour : $[Fe(H_2O)_6]^{2+}$
   (ii) Write the IUPAC name of the coordination complex : $[CoCl_2(en)_2]NO_3$

   OR

   (i) Predict the geometry of $[Ni(CN)_4]^{2-}$.
   (ii) Calculate the spin only magnetic moment of $[Cu(NH_3)_4]^{2+}$ ion. *(2020-21)*

SA II (3 marks)

7. Using Valence bond theory, explain the following in relation to the paramagnetic complex $[Mn(CN)_6]^{3-}$
   (a) type of hybridisation
   (b) magnetic moment value
   (c) type of complex – inner, outer orbital complex
   (2022-23)

8. Answer the following questions :
   (a) $[Ni(H_2O)_6]^{2+}_{(aq)}$ is green in colour whereas $[Ni(H_2O)_4(en)]^{2+}_{(aq)}$ is blue in colour, give reason in support of your answer.
   (b) Write the formula and hybridisation of the following compound :
   tris(ethane-1,2-diamine)cobalt(III) sulphate U

   OR

   In a coordination entity, the electronic configuration of the central metal ion is $t_{2g}^3 e_g^1$.
   (a) Is the coordination compound a high spin or low spin complex?
   (b) Draw the crystal field splitting diagram for the above complex. *(Term II, 2021-22)*

Detailed SOLUTIONS

Previous Years' CBSE Board Questions

1. (c) : $[Co(NH_3)_5SO_4]Cl \xrightarrow{AgNO_3} \underset{\text{(white ppt)}}{AgCl\downarrow}$

Only Cl^- ion gets ionise, SO_4^{2-} will not ionise since it is present inside coordination sphere.

2. (a): For one mole of the compound, two moles of AgCl are precipitated which indicates two ionisable chloride ions in the complex. Hence, its structural formula is $[CrCl(H_2O)_5]Cl_2 \cdot H_2O$

3. For one mole of the compound, two moles of AgCl are precipitated which indicates two ionisable chloride ions in the complex. Hence, its structural formula is $[CrCl(H_2O)_5]Cl_2 \cdot H_2O$

Commonly Made Mistake

- Student must know the difference between primary and secondary valency.
- Primary valency - Normally ionisable and are satisfied by negative ions.
- Secondary valency - Non-ionisable and are satisfied by neutral molecules or negative ions.

4. Double salts dissociate into ions completely when dissolved in water. On the other hand, in complexes, the complex ion does not dissociate.

5. Structural formula : $[Co(NH_3)_6]Cl_3$

6. (b): $[Co(en)_3]_2(SO_4)_3$
Let oxidation number of Co be *x*.
$2(+x + 3(0)) = +6$
$2x = +6$
$x = +3$

7. (c): $C_2O_4^{2-}$ is a chelating ligand, it forms a chelate complex with metal ion.

8. (a): The coordination number of Cr in $[CrCl_2(ox)_2]^{3-}$ is 6 as Cl is monodentate ligand and oxalate is bidentate ligand.

9. (b): Ethane-1, 2-diamine (*en*) is a bidentate ligand.

Key Points

- The secondary valency is equal to the coordination number.

10. (d): Formation of chelate rings increases the stability of the complex.

11. (a): As CO is a neutral ligand, its oxidation state is zero. The oxidation state of Ni in $Ni(CO)_4$ is calculated as follows :
$x + 4(0) = 0, x = 0$

12. $[Fe(C_2O_4)_3]^{3-}$ is more stable than $[Fe(NH_3)_6]^{3+}$ due to chelate effect, as it forms rings.

13. Coordination number and oxidation state of Pt in the complex $[Pt(en)_2Cl_2]$ are 6 and +2 because *en* is a bidentate and neutral ligand.

14. $[Co(en)_3]^{3+}$ is more stable complex than $[Co(NH_3)_6]^{3+}$ due to chelate effect as it forms rings.

15. (c) : $[Co(NH_3)_4Cl(NO_2)]Cl$
Tetraamminechloridonitrito-N-cobalt(III) chloride

16. $Fe_4[Fe(CN)_6]_3$

17. IUPAC name of the complex $[Cr(NH_3)_3Cl_3]$ is triamminetrichloridochromium(III).

18. Pentaamminecarbonatocobalt(III) chloride

19. IUPAC name of the complex
For one mole of the compound, two moles of AgCl are precipitated which indicates two ionisable chloride ions in the complex. Hence, its structural formula is

$[CrCl(H_2O)_5]Cl_2 \cdot H_2O$
$[CrCl(H_2O)_5]Cl_2 \cdot H_2O$ is :
Pentaaquachloridochromium(III) chloride.

20. (a) $K_2[Zn(OH)_4]$
Potassium tetrahydroxidozincate(II)
(b) $[Pt(NH_3)_6]Cl_4$
Hexaammineplatinum(IV) chloride

Key Points

- If the complex provides anionic complex ion then the name of the metal ends with the suffix *ate*.

21. (i) $[Co(NH_3)_5(ONO)]Cl_2$ (ii) $K_2[Ni(CN)_4]$

22. (a) $Na[Au(CN)_2]$
Sodium dicyanidoaurate(I)
(b) $[Pt(NH_3)_4(Cl)(NO_2)]SO_4$
Tetraamminechloridonitrito-N-platinum(IV) sulphate

23. (a) $[Cr(en)_3]Cl_3$
Tris(ethane-1,2-diamine)chromium(III) chloride
(b) $K_2[Zn(OH)_4]$
Potassium tetrahydroxidozincate(II)

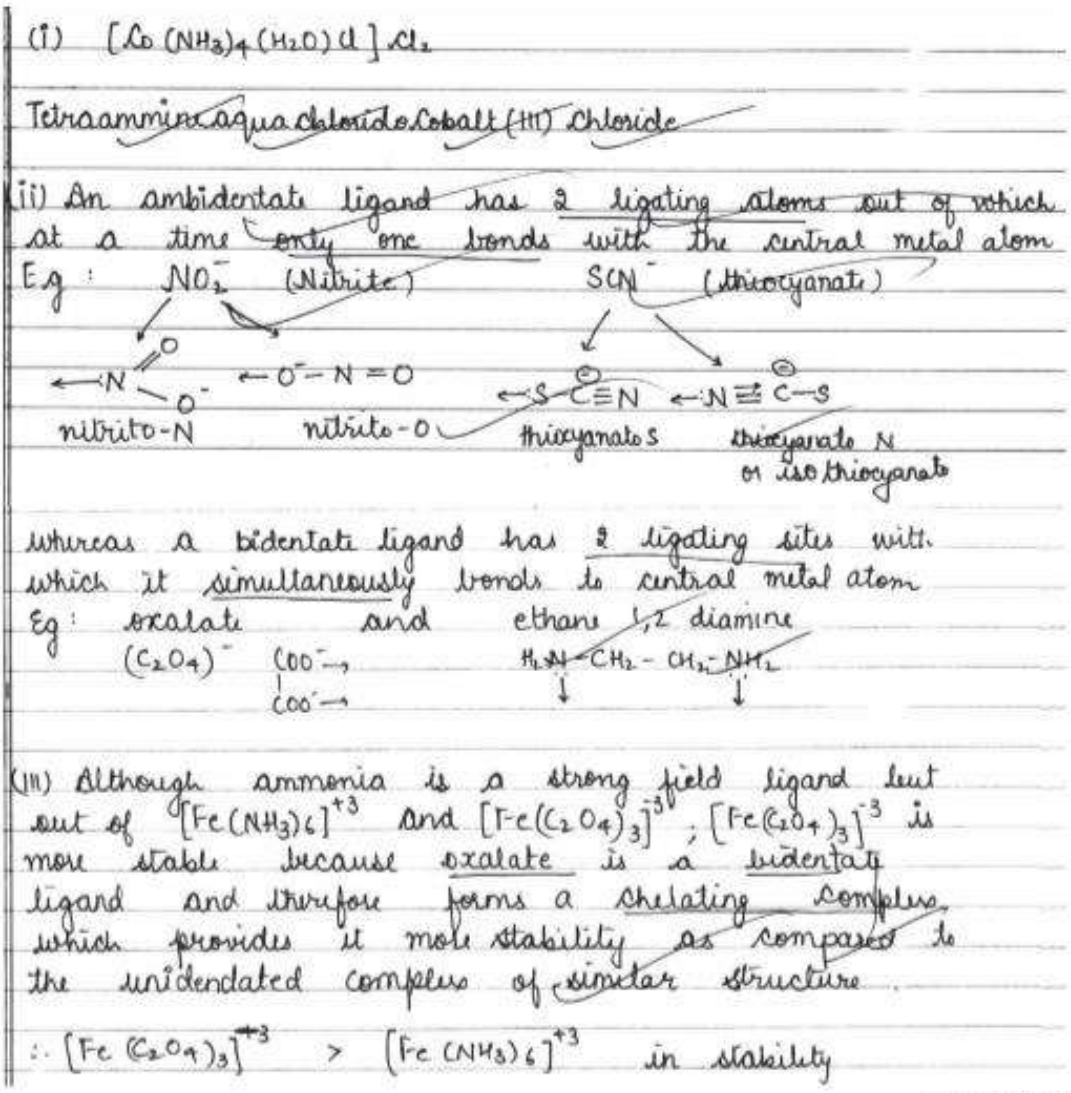
24. (a) $K_3[Al(C_2O_4)_3]$
Potassium trioxalatoaluminate(III)
(b) $[CoCl_2(en)_2]^+$
Dichlorido*bis*(ethane-1,2-diamine)cobalt(III) ion

25. (i) Diamminedichlorido(ethane-1,2-diamine) chromium(III) chloride.
(ii) $[Co(NH_3)_5(ONO)]^{2+}$

26. (i) $[Co(NH_3)_6]Cl_3$
(ii) $K_2[NiCl_4]$

27. (i) *Tris*(ethylenediammine)chromium(III) chloride
(ii) $K_3[Cr(ox)_3]$

28.

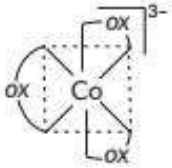


[Topper's Answer, 2022]

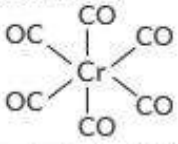
29. (i) Hexaamminecobalt(III) chloride
(ii) Tetrachloridonickelate(II) ion
(iii) Potassium hexacyanidoferrate(III)

30. (i) Pentaamminechloridocobalt(III) chloride
(ii) Potassium hexacyanidoferrate(III)
(iii) Tetrachloridonickelate(II) ion

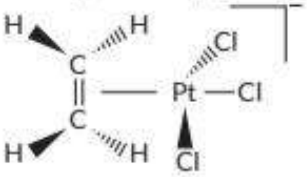
31. (i) Trioxalatocobaltate(III)



(ii) Hexacarbonylchromium(0)



(iii) Trichloridoetheneplatinate(II) ion



32. (c) : Compounds, which give different ions in solution due to the exchange of ions in coordination sphere and counter ions are called ionisation isomers.
Therefore, $[Co(Br)(NH_3)_5]SO_4$ and $[Co(SO_4)(NH_3)_5]Br$ are ionisation isomers.

33. (c) : The pair $[Co(NH_3)_4Cl_2]Br_2$ and $[Co(NH_3)_4Br_2]Cl_2$ will show ionisation isomerism.

34. (a)

35. Linkage isomerism is shown by the given complex as NO_2 is an ambidentate ligand. The resultant linkage isomers are $[Co(NH_3)_5(NO_2)]Cl_2$ and $[Co(NH_3)_5(ONO)]Cl_2$. Ionisation isomerism is also shown by the given complex as $[Co(NH_3)_5(NO_2)]Cl_2$ and $[Co(NH_3)_5Cl]NO_2Cl$

36. Ionisation isomerism : $[Co(NH_3)_5Cl]SO_4$ and $[Co(NH_3)_5SO_4]Cl$

37. Coordination isomerism

38. The complex $[Co(NH_3)_5(SCN)]^{2+}$ shows linkage isomerism as SCN^- is an ambidentate ligand.

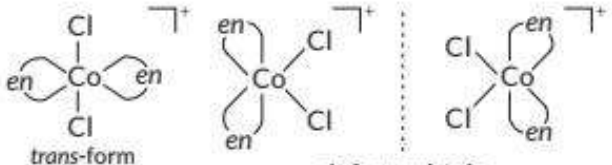
39. Linkage isomerism : $[Co(NH_3)_5NO_2]^{2+}$ and $[Co(ONO)(NH_3)_5]^{2+}$

Concept Applied

➲ Linkage isomerism arises in a coordination compound containing ambidentate ligand.

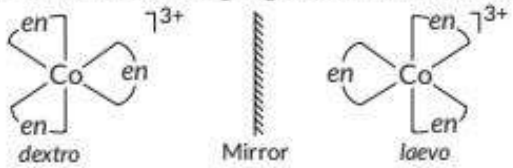
40. (a) An atom, molecule or ion which is linked to central metal atom or ion through coordinate bond by donating lone pair of electrons present on its donor atom is known as ligand.
Here, NH_4^+ cannot act as a ligand since there is no lone pair of electrons on its donor atom.
(b) Linkage isomer of $[Co(NH_3)_5(NO_2)]Cl_2$ is $[Co(NH_3)_5(ONO)]Cl_2$
IUPAC name : Pentaamminenitrito-O-cobalt(III) chloride.

41. $[CoCl_2(en)_2]^+$: Dichlorido*bis*(ethane-1,2-diamine) cobalt(III) ion
This complex exhibit both optical and geometrical isomerism and *cis* isomer is optically active.

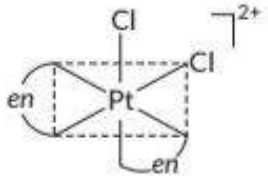


cis-form exists in two enantiomeric forms (*d* and *l*)

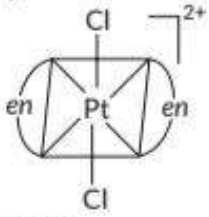
42. The complex, $[Co(en)_3]Cl_3$ shows optical isomerism.



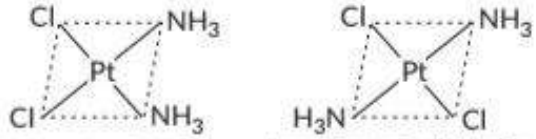
43. *Cis*-isomer of the complex $[Pt(en)_2Cl_2]^{2+}$ is optically active.



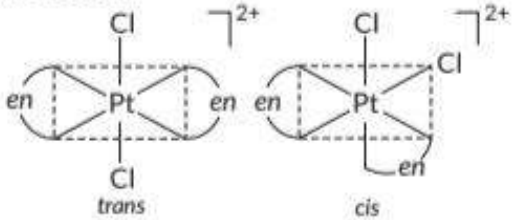
44. *Trans*-isomer is optically inactive due to the presence of plane of symmetry.



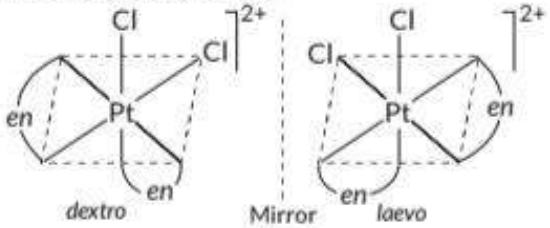
45. Geometrical isomers



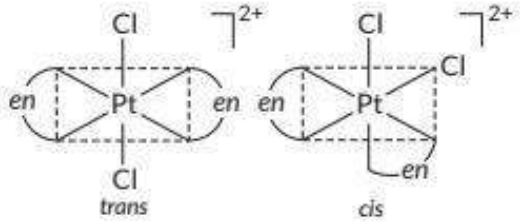
46. IUPAC name - Dichlorido*bis*(ethane-1,2-diamine) platinum(IV) ion. This complex shows geometrical and optical isomerism.



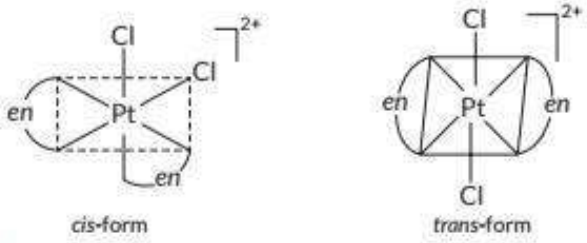
Cis-isomer is optically active.



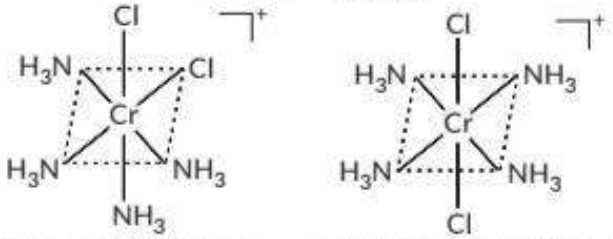
47. This complex shows geometrical and optical isomerism. Geometrical isomers are *cis*-*trans* isomers.



48. $[Pt(en)_2Cl_2]$
Dichloridodiethylenediamine platinum(II)
Geometrical isomers of $[Pt(en)_2Cl_2]$ are

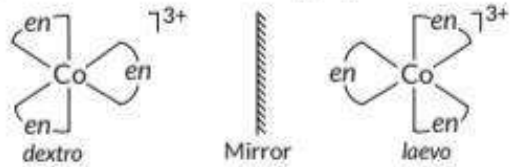


49. Tetraamminedichloridochromium(III) ion
Geometrical isomers of $[Cr(NH_3)_4Cl_2]^+$;

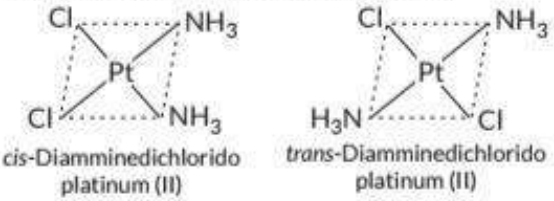


50. (i) Linkage isomerism : $[Co(NH_3)_5NO_2]^{2+}$ and $[Co(NH_3)_5(ONO)]^{2+}$

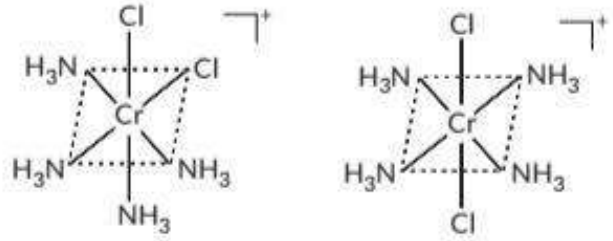
(ii) Optical isomerism : $[Co(en)_3]Cl_3$



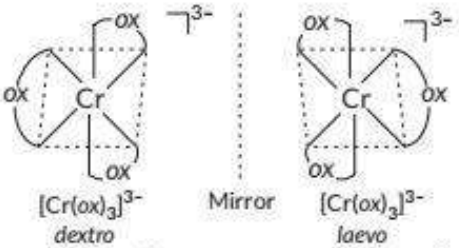
(iii) Geometrical isomerism : $[Pt(NH_3)_2Cl_2]$



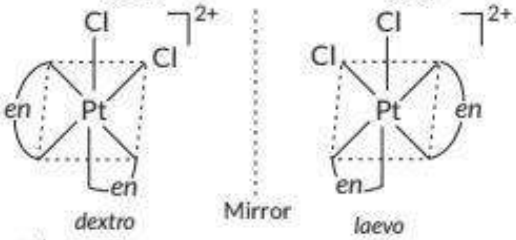
51. Tetraamminedichloridochromium(III) ion $[Cr(NH_3)_4Cl_2]^+$ shows geometrical isomerism.



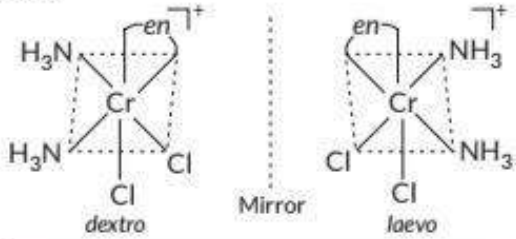
52. $[Cr(C_2O_4)]^{3-}$: ox



$[PtCl_2(en)_2]^{2+}$:



$[Cr(NH_3)_2Cl_2(en)]^+$:

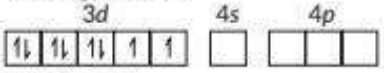


Key Points

- Optical isomers are mirror images that cannot be superimposed on one another.

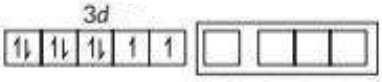
53. (b): In $[NiCl_4]^{2-}$, Ni is in +2 oxidation state and has $3d^8$ outer electronic configuration .

Orbital of Ni^{2+} ion

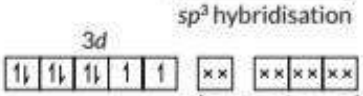


Since, Cl^- is a weak field ligand.

sp^3 hybridised orbital of Ni^{2+}



$[NiCl_4]^{2-}$
high spin complex



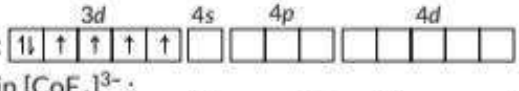
It is sp^3 hybridised high spin paramagnetic complex.

$\mu = \sqrt{n(n+2)}$

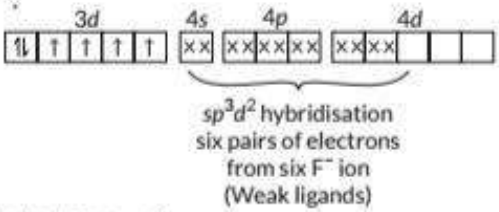
$= \sqrt{2(2+2)} = \sqrt{2\times4} = \sqrt{8} = 2{\cdot}82$ B.M.

54. The IUPAC name of $[CoF_6]^{3-}$ is hexafluoridocobaltate(III).
Oxidation state of Co ion in $[CoF_6]^{3-}$ is +3.

Co^{3+} :



Co^{3+} in $[CoF_6]^{3-}$:



No. of unpaired electrons = 4

Key Points

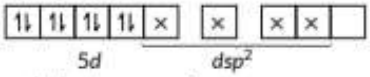
- Weak field ligands give, outer orbital or high spin or spin free complex.

55. $[Pt(CN)_4]^{2-}$ has square planar structure.

Pt : $[Xe]\,4f^{14}\,5d^9\,6s^1$

Pt^{2+} : $[Xe]\,4f^{14}\,5d^8\,6s^0$

$[Pt(CN)_4]^{2-}$:

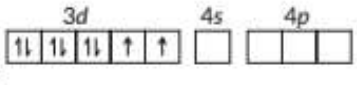


No. of unpaired electrons = 0.

56. The complex $[Co(C_2O_4)_3]^{3-}$ has d^2sp^3-hybridisation and it is diamagnetic in nature due to the absence of unpaired electrons.

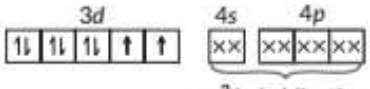
57. $[NiCl_4]^{2-}$ contains Ni^{2+} ion with $3d^8$ configuration.

Ni^{2+} : Ground state



Cl^- is a weak field ligand thus no pairing will occur. Hence, outer 4s and 4p-orbitals are used in hybridisation.

$[NiCl_4]^{2-}$: with $4Cl^-$ ligands

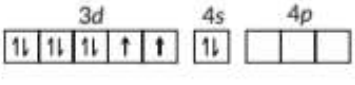


sp^3 hybridisation, (tetrahedral) 4 electron pairs donated by $4Cl^-$ ligands (weak field ligand)

It has two unpaired electrons hence, it is paramagnetic.

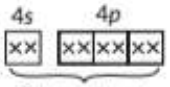
$[Ni(CO)_4]$ contains Ni(0) – $3d^84s^2$ configuration.

Ni(0) : Ground state



CO is a strong field ligand hence, 4s-electrons will shift to 3d-orbital making 4s-orbital vacant.

$[Ni(CO)_4]$:



sp^3 hybridisation (tetrahedral) four electrons donated by four CO ligands (strong field ligand)

The complex has all paired electrons hence, it is diamagnetic.

Key Points

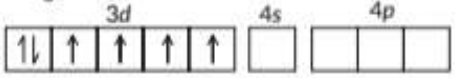
- Strong field ligands give, inner orbital or low spin or spin paired complex. Weak field ligands give, outer orbital or high spin or spin free complex.

58. (i) $[Mn(H_2O)_6]SO_4$

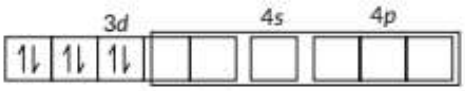
Hexaaquamanganese(II) sulphate

(ii) $[Fe(CN)_6]^{4-}$ ion :

Fe^{2+} ion :

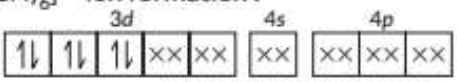


Fe^{2+} ion hybridised (under the influence of strong field ligand).



Six empty d^2sp^3 hybrid orbitals

$[Fe(CN)_6]^{4-}$ ion formation :

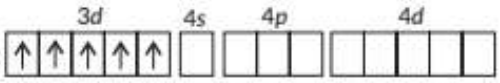


Six pairs of electrons from six CN^- ions (d^2sp^3)

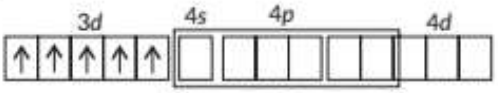
Since, the complex ion does not contain any unpaired electron, so it is diamagnetic.

$[FeF_6]^{3-}$ ion :

Fe^{3+} ion :

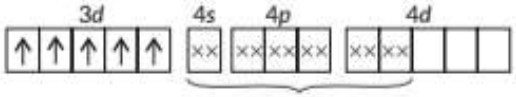


Fe^{3+} ion hybridised (under the influence of weak field ligand).



Six empty sp^3d^2 hybrid orbitals

$[FeF_6]^{3-}$ ion formation :



Six electrons pairs from six F^- ions (sp^3d^2)

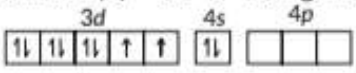
As the complex ion contains five unpaired electrons, it is highly paramagnetic in nature.

$\mu_s = \sqrt{5(5+2)} = \sqrt{35} = 5.9$ B.M.

59. (i) IUPAC name; Tetracarbonylnickel(0)

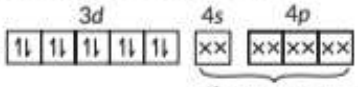
$[Ni(CO)_4]$ contains Ni(0) – $3d^84s^2$ configuration.

Ni(0) : Ground state



CO is a strong field ligand hence, 4s-electrons will shift to 3d-orbital making 4s-orbital vacant.

$[Ni(CO)_4]$:

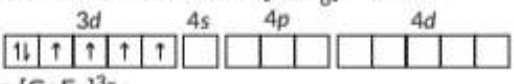


sp^3 hybridisation (tetrahedral) four electrons donated by four CO ligands (strong field ligand)

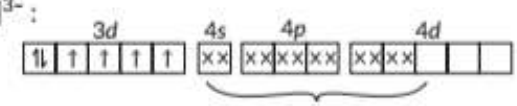
(ii) The IUPAC name of $[CoF_6]^{3-}$ is hexafluoridocobaltate(III).

Oxidation state of Co ion in $[CoF_6]^{3-}$ is +3.

Co^{3+} :



Co^{3+} in $[CoF_6]^{3-}$:



sp^3d^2 hybridisation six pairs of electrons from six F^- ion (Weak field ligands)

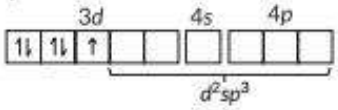
60. (a) $[Fe(CN)_6]^{3-}$

IUPAC name : Hexacyanidoferrate (III) ion

Fe is present as Fe^{3+} in $[Fe(CN)_6]^{3-}$

Fe^{3+} : [Ar] $3d^5\,4s^0$

As CN^- is a strong field ligand, hence, electrons in 3*d* get paired up.

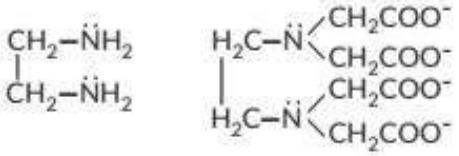


Hybridisation is d^2sp^3.

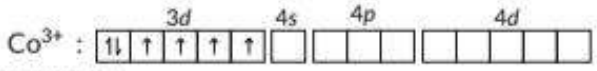
(b) Ambidentate ligands : These are the unidentate ligands containing more than one coordinating atoms are called ambidentate ligands.

For example NO_2; it can donate through –N or –O both.

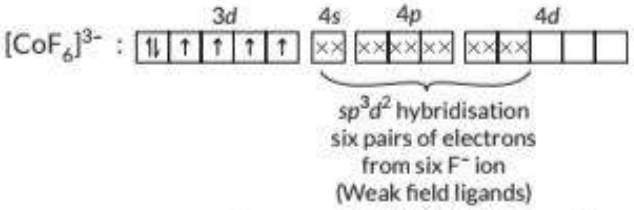
Chelating ligands : Bidentate or polydentate ligands which contain two or more donor atoms and these are positioned in such a way that a five or six membered ring is formed with metal ion. for example ethylenediamine and $EDTA^-$.

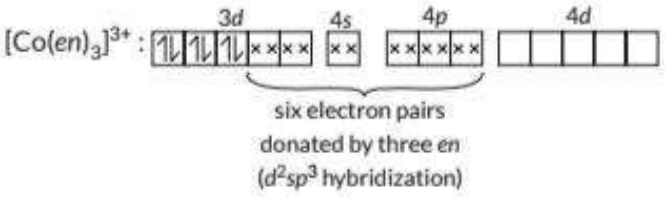


61. Oxidation state of Co ion in $[CoF_6]^{3-}$ is +3.



(In ground state)





(i) $[CoF_6]^{3-}$ is paramagnetic in nature.

(ii) Chelating ligands form more stable cyclic/ring complexes than unidentate or non-chelating ligands. Since, $[Co(en)_3]^{3+}$ contains *en* which is a bidentate chelating ligand, it is more stable than $[CoF_6]^{3-}$.

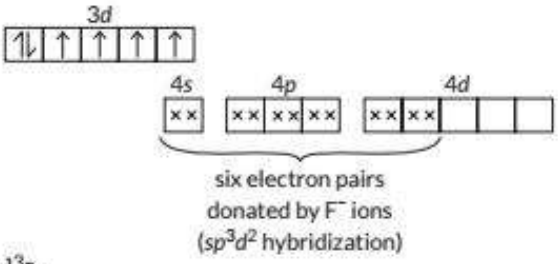
(iii) $[Co(en)_3]^{3+}$ is an inner orbital complex as its hybridisation is d^2sp^3 hybridisation.

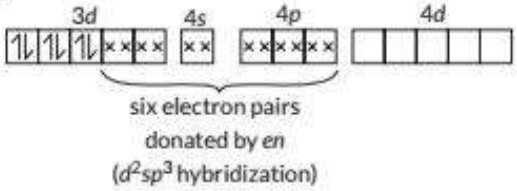
(iv) $[CoF_6]^{3-}$ is a high spin complex.

Concept Applied

➲ Strong field ligand forms low spin, inner orbital complexes and weak field ligands from high spin outer orbital complex.

62. Formation of $[CoF_6]^{3-}$ and $[Co(C_2O_4)_3]^{3-}$ can be represented as :





(i) $[Co(C_2O_4)_3]^{3-}$ is diamagnetic as all electrons are paired.

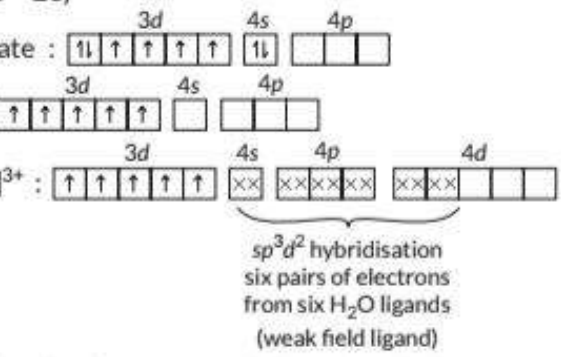
(ii) $[Co(C_2O_4)_3]^{3-}$ is more stable as $C_2O_4^{2-}$ is a chelating ligand and forms chelate rings.

(iii) $[CoF_6]^{3-}$ is outer orbital complex as it undergoes sp^3d^2 hybridization using the outer 4*d*-orbital.

(iv) $[Co(C_2O_4)_3]^{3-}$ is low spin complex due to absence of any unpaired electron.

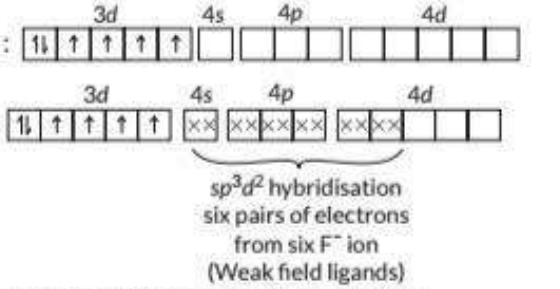
63. $[Fe(H_2O)_6]^{3+}$:

Fe atom (Z = 26)

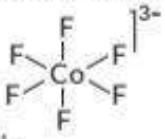


The complex ion has outer orbital octahedral geometry (high spin) and is paramagnetic due to the presence of five unpaired electrons.

64. (i) Oxidation state of Co ion in $[CoF_6]^{3-}$ is +3.

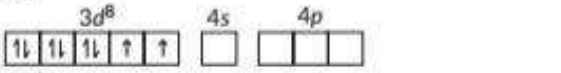


Structure – Outer orbital octahedral complex



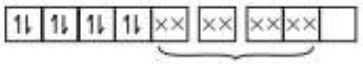
Nature – Paramagnetic

(ii) In $[Ni(CN)_4]^{2-}$: Ni is present as Ni(II) with $3d^8$ configuration.



(In ground state)

$[Ni(CN)_4]^{2-}$:

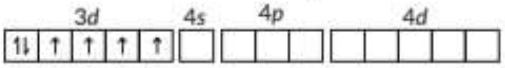


dsp^2 hybridisation
four electrons pairs
donated by four
CN^- ions
(Strong field ligand)

The complex ion has square planar geometry and is diamagnetic in nature.

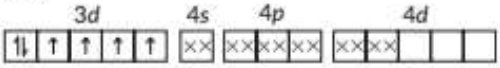
65. Oxidation state of Co ion in $[CoF_6]^{3-}$ is +3.

Co^{3+} :



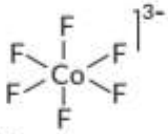
(In ground state)

$[CoF_6]^{3-}$:



sp^3d^2 hybridisation
six pairs of electrons
from six F^- ion
(Weak field ligands)

Structure – Outer orbital octahedral complex



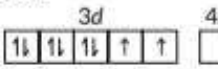
Nature – Paramagnetic
IUPAC Name : Hexafluoridocobaltate(III) ion

Key Points

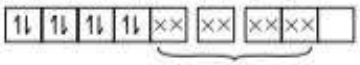
➲ F^- is a weak field ligands.

66. In $[Ni(CN)_4]^{2-}$: Ni is present as Ni(II) with $3d^8$ configuration.

Ni^{2+} :



$[Ni(CN)_4]^{2-}$:

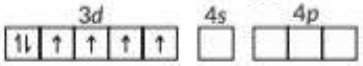


dsp^2 hybridisation
four electrons pairs
donated by four
CN^- ions
(Strong field ligand)

The complex ion has square planar geometry and is diamagnetic in nature.
IUPAC name : Tetracyanidonickelate(II) ion

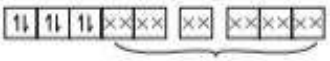
67. Oxidation of cobalt in $[Co(NH_3)_6]^{3+}$ is +3.

Co^{3+} :



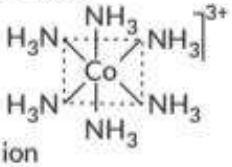
Ground State

$[Co(NH_3)_6]^{3+}$ ion :



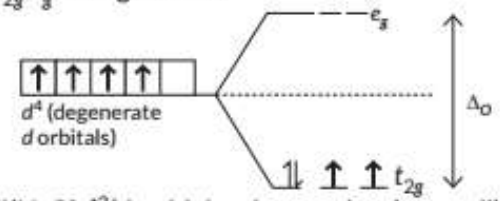
d^2sp^3 hybridisation
six pairs of electrons
from six NH_3 ligand
(Strong field ligand)

Hybridisation – d^2sp^3
Structure – Octahedral
(low spin)



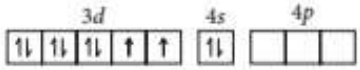
Nature – Diamagnetic
IUPAC name : Hexaamminecobalt(III) ion

68. (a) On the basis of CFT, if strong field ligand is present for d^4 orbit with $\Delta_o > P$, the energy gap between t_{2g} and e_g will be more and thus the electron will pair up giving $t_{2g}^4 e_g^0$ configuration.



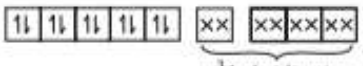
(b) $[Ni(H_2O)_6]^{2+}$ is a high spin complex (Δ_o small).
In $[Ni(H_2O)_6]^{2+}$, d-d transitions are taking place on absorbing low energy radiation (red component of spectrum) from visible region showing green as the complementary colour.
$[Ni(CO)_4]$ contains Ni(0) – $3d^8 4s^2$ configuration.

Ni(0) :



Ground state
CO is a strong field ligand hence, 4s-electrons will shift to 3d-orbital making 4s-orbital vacant.

$[Ni(CO)_4]$:

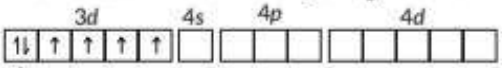


sp^3 hybridisation
(tetrahedral)
four electrons
denoted by four
CO ligands
(strong ligand)

The complex has all paired electrons hence, it is colourless.

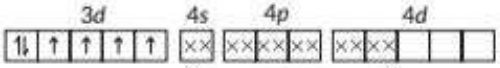
69. (a) Oxidation state of Co ion in $[CoF_6]^{3-}$ is +3.

Co^{3+} :



(In ground state)

$[CoF_6]^{3-}$:



sp^3d^2 hybridisation
six pairs of electrons
from six F^- ion
(Weak field ligands)

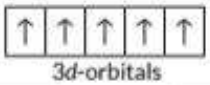
Structure – Outer orbital octahedral complex



Nature – Paramagnetic
(b) $[Co(NH_3)_5Cl]Cl$
Pentaamminechloridocobalt(III) chloride

70. (i) $[FeF_6]^{3-}$
Oxidation state of Fe = +3

Fe^{3+} : $3d^5$,



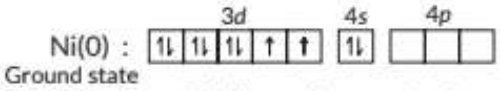
3d-orbitals

$\mu_{spin} = \sqrt{n(n+2)}$ B.M. $= \sqrt{5(5+2)}$ B.M $= 5.91$ B.M.

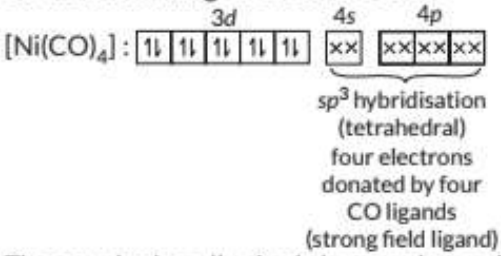
(ii) Pentaamminechloridocobalt(III) chloride
(iii) Chelating ligands form more stable cyclic/ring complexes than unidentate or non-chelating ligands.

Since, $[Co(en)_3]^{3+}$ contains *en* which is a bidentate chelating ligand, it is more stable than $[CoF_6]^{3-}$.

71. (a) $[Ni(CO)_4]$ contains Ni(0) – $3d^8 4s^2$ configuration.



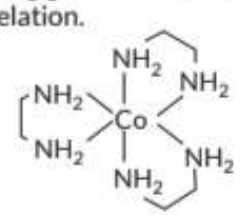
CO is a strong field ligand hence, 4*s*-electrons will shift to 3*d*-orbital making 4*s*-orbital vacant.



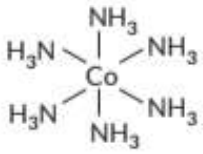
The complex has all paired electrons hence, it is diamagnetic.

(b) Diamminechloridonitrito-N-platinum(II)

(c) $[Co(en)_3]^{3+}$ is a complex having bidentate ligand (*en*) that form a five - membered ring around Co metal that results in chelation. While no such ring is formed in $[Co(NH_3)_6]^{3+}$ complex. So, $[Co(en)_3]^{3+}$ is more stable due to chelation.



(*en* = ethylenediamine)

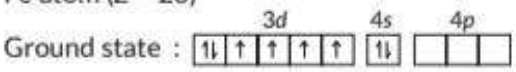


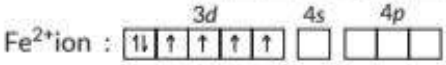
Key Points

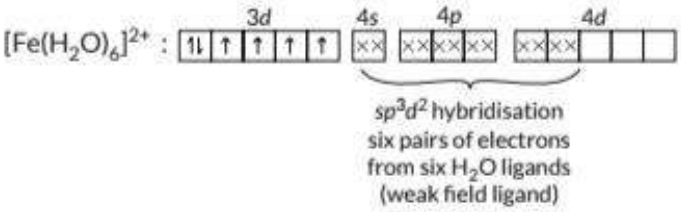
➲ Chelating ligands form more stable complexes.

72. (i) $[Fe(H_2O)_6]^{2+}$:

Fe atom (Z = 26)

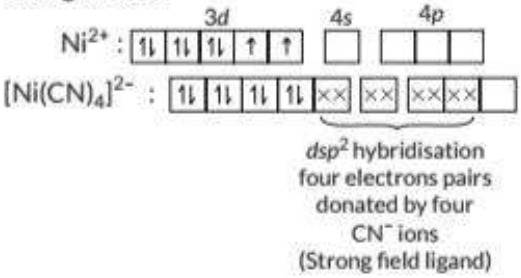






The complex ion has outer orbital octahedral geometry (high spin) and is paramagnetic due to the presence of four unpaired electrons.

(ii) In $[Ni(CN)_4]^{2-}$: Ni is present as Ni(II) with $3d^8$ configuration.



The complex ion has square planar geometry and is diamagnetic in nature.

73. (i) Its formula will be $[Pt(NH_3)_2Cl_2]$.

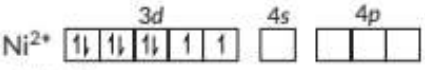
(ii) *en* is a bidentate ligand. Hence in $[Co(en)_3]^{3+}$, its secondary valency is 6.

(iii) (1) $Fe_4[Fe(CN)_6]_3$ is Iron (III) hexacyanidoferrate(II).

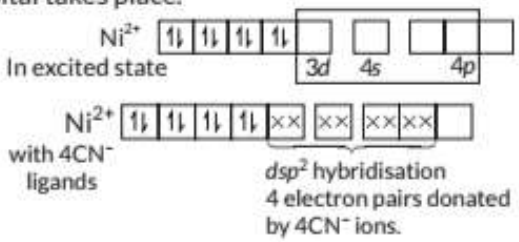
(2) IUPAC name of $[Co(NH_3)_5Cl]Cl_2$ is pentaammine chloridocobalt(III) chloride.

OR

(iii) In $[Ni(CN)_4]^{2-}$, Ni is present as Ni(II) with $3d^8$-configuration.



CN^- is a strong field ligand. Hence pairing of electrons in 3*d*-orbital takes place.



It has all paired electrons, hence it is diamagnetic.

74. (a): If $\Delta_o < P$ (*i.e.*, Δ_o small, weak field ligand) → electrons spread out among all *d*-orbitals before pairing up → forms high spin complex.

For tetrahedral crystal field, the order of splitting is reversed and is related to Δ_o as

$$\Delta_t = \frac{4}{9}\Delta_o$$

Consequently the orbital splitting energies are not sufficiently large for forcing pairing and therefore, low spin configurations are rarely observed in tetrahedral coordination entities.

75. (c)

76. $[Ni(H_2O)_6]^{2+}$ is a high spin complex (Δ_o small) while $[Ni(CN)_4]^{2-}$ is a low spin square planar complex.

In $[Ni(H_2O)_6]^{2+}$ complex, *d-d* transitions are taking place on absorbing low energy radiation (red component of spectrum) from visible region showing green as the complementary colour.

In $[Ni(CN)_4]^{2-}$ complex, *d-d* transitions do not take place because there is no unpaired electron present hence, complex is colourless.

77. (a) Due to the presence of weak field ligand, *i.e.*, Cl^- in the complex $[NiCl_4]^{2-}$ two unpaired electrons are present in 3*d*-orbitals of Ni-atom hence, this complex is paramagnetic in nature. On the other hand, due to the presence of strong field ligand *i.e.*, CN^- in the complex $[Ni(CN)_4]^{2-}$, no unpaired electron is present in 3*d*-orbitals of Ni-atom (as strong field ligand causes pairing of electrons), hence, it is diamagnetic in nature.

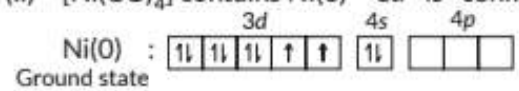
(b) In tetrahedral coordination entity formation, the *d*-orbital splitting is smaller. Consequently, the orbital splitting energies are not sufficiently large to force

pairing and therefore, low spin configurations are rarely observed.

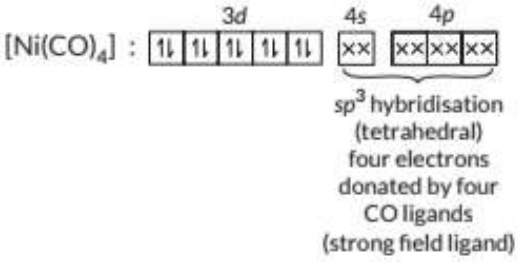
78. (i) For d^4 ion, if $\Delta_o < P$, the fourth electron enters one of the e_g orbitals giving the configuration $t_{2g}^3 e_g^1$.

Ligands for which $\Delta_o < P$ are known as weak field ligands and form high spin complexes.

(ii) $[Ni(CO)_4]$ contains Ni(0) – $3d^8 4s^2$ configuration.



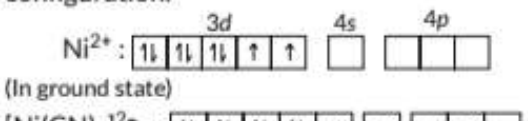
CO is a strong field ligand hence, 4*s*-electrons will shift to 3*d*-orbital making 4*s*-orbital vacant.



The complex has all paired electrons hence, it is diamagnetic.

79. (i) When $\Delta_o < P$, $t_{2g}^3 e_g^1$

(ii) In $[Ni(CN)_4]^{2-}$: Ni is present as Ni(II) with $3d^8$ configuration.



The complex ion has square planar geometry and is diamagnetic in nature.

(iii) $[Co(en)_2Cl_2]^+$

Dichlorido*bis*(ethane-1,2-diamine)cobalt(III) ion

80. (i) For d^4 ion, if $\Delta_o < P$, the fourth electron enters one of the e_g orbitals giving the configuration $t_{2g}^3 e_g^1$.

Ligands for which $\Delta_o < P$ are known as weak field ligands and form high spin complexes.

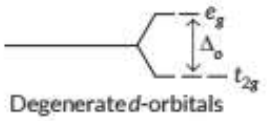
(ii) Due to the presence of weak field ligand, *i.e.*, Cl^- in the complex $[NiCl_4]^{2-}$ two unpaired electrons are present in 3*d*-orbitals of Ni-atom hence, this complex is paramagnetic in nature. On the other hand, due to the presence of strong field ligand *i.e.*, CN^- in the complex $[Ni(CN)_4]^{2-}$, no unpaired electron is present in 3*d*-orbitals of Ni-atom (as strong field ligand causes pairing of electrons), hence, it is diamagnetic in nature.

(iii) $[PtCl_2(NH_3)_4]Cl_2 \rightarrow [PtCl_2(NH_3)_4]^{2+}_{(aq)} + 2Cl^-_{(aq)}$

Hence, the total number of ions produced is three.

81. (a) $K_3[Cr(C_2O_4)_3]$: Potassium trioxalatochromate(III)

(b) According to crystal field theory, five *d*-orbitals, split into two different energy levels termed as t_{2g} and e_g as shown below :



Degenerate *d*-orbitals

For d^5 ion with $\Delta_o < P$, electronic configuration will be $t_{2g}^3 e_g^2$.

Concept Applied

➲ Weak field ligands causes less splitting, for these $\Delta_o < P$, hence before pairing electron goes to higher e_g level, causing formation of high spin complexes.

(c) Ambidentate ligand : A unidentate ligand which can coordinate to central metal atom through two different atoms is called ambidentate ligand. For example, NO_2^- ion can coordinate either through nitrogen or through oxygen to the central metal atom/ion.

82. (i) $K_2[PdCl_4]$: Potassium tetrachloridopalladate(II)

(ii) If $\Delta_o > P$, it becomes energetically favourable for the fourth and fifth electron to occupy in t_{2g} orbital with the resultant configuration $t_{2g}^5 e_g^0$. Such an effect is produced by strong field ligand and such a complex formed is referred to as low spin complex.

(iii) Homoleptic complexes : Complexes in which a metal is bond to only one kind of ligand are called homoleptic complexes.

e.g., $[Co(NH_3)_6]^{3+}$, $[Cu(CN)_6]^{3-}$, $[Ti(H_2O)_6]^{3+}$

83. (i) The complexes formed by chelating ligand are more stable than that formed by unidentate ligand because of multiple points of attachment of the ligand with the central atom/ion and formation of stable cyclic closed ring structures.

(ii) A spectrochemical series is an experimentally determined series based on the absorption of light by complexes with different ligands. It contains an arrangement of ligands in increasing order of their field strength. The series can be listed as :

$I^- < Br^- < SCN^- < Cl^- < S^{2-} < F^- < OH^- < C_2O_4^{2-} < H_2O < NCS^- < edta^{4-} < NH_3 < en < CN^- < CO$

A strong field ligand produces a stronger field and causes greater splitting of the *d*-orbitals. In this case, pairing of electrons occurs. For example : CO, CN^-, etc. On the other hand, a weak field ligand produces a weak field and causes lesser splitting of *d*-orbitals. In such case, pairing of electrons does not occur. For example : Cl^-, Br^-, I^-, etc.

84. (a) $K_4[Mn(CN)_6]$

Potassium Hexacyanomanganate(II)

In $K_4[Mn(CN)_6]$, Mn is present in +2 oxidation state.

Mn^{2+} : d^5 configuration

In presence of strong ligand, $\Delta_o < P$

Hence : $t_{2g}^5 e_g^0$

(b) A ligand which forms two or more bonds with central atom or ion and results in the formation of stable cyclic complex is called chelating ligand and the effect is called chelating effect. For example, oxalate, ethylene diamine, dimethyl glyoxime, etc.

85. (a) Ambidentate ligand : Ligands having two different atoms through which it can act as a ligand

For example,

CN^- CN^- : $\overset{\leftarrow}{\cdot\cdot}C\equiv N$ or $\ddot{C}=\dot{N}:\rightarrow$

Cyanido-C Cyanido-N

(b) A spectrochemical series is an experimentally determined series based on the absorption of light by complexes with different ligands. It contains an arrangement of ligands in increasing order of their field strength. The series can be listed as :

$I^- < Br^- < SCN^- < Cl^- < S^{2-} < F^- < OH^- < C_2O_4^{2-} < H_2O < NCS^- < edta^{4-} < NH_3 < en < CN^- < CO$

(c) Heteroleptic complexes : Complexes in which the central atom is bound to different type of ligands are called *heteroleptic complexes.*

e.g., $[Co(NH_3)_4Cl_2]$, $K_2[Fe(CN)_5NO]$, $[Fe(H_2O)_5NO]SO_4$

86. In CO, both lone pair of electrons and vacant π^* orbitals are present. Hence, it acts as electron pair σ donor as well as π acceptor by back bonding. Hence, M—CO bond is stronger.

$$M \underset{\sigma}{\overset{\pi}{\rightleftharpoons}} CO$$

NH_3 is electron pair donor only. Accumulation of negative charge on the metal ion takes place, hence M—NH_3 bond is weaker.

$M \leftarrow NH_3$

87. (c) : Hardness of water is estimated by titration with EDTA. EDTA is a hexadentate ligand.

CBSE Sample Questions

1. (c) : 1 mole of Mohr's salt gives 1 Fe^{2+}, 2SO_4^{2-} and 2NH_4^+ ions on dissolving in water. The total number of ions is 5. **(1)**

2. (a): Ambidentate ligands such as NO_2^-, SCN^- are unidentate ligands. **(1)**

3. (a): Tetraammineaquachloridocobalt(III) chloride : $[Co(NH_3)_4(H_2O)Cl]Cl_2$ **(1)**

4. $[Co(NH_3)_5CO_3]Cl$ and $[Co(NH_3)_5Cl]CO_3$

Pentaamminecarbonatocobalt(III) chloride

Pentaamminechloridocobalt(III) carbonate **(2)**

5. (b): Using equation, $\Delta_t = \frac{4}{9}\Delta_o$

$\Delta_t = (4/9) \times 18000\ cm^{-1} = 8000\ cm^{-1}$ **(1)**

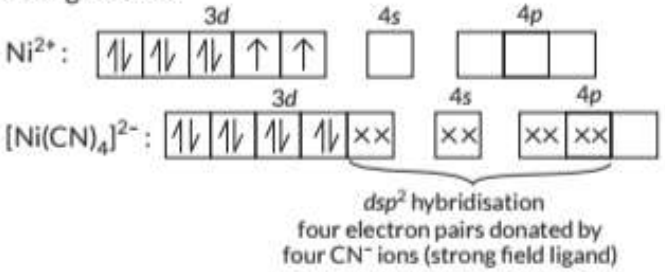
6. (i) Electronic configuration : $t_{2g}^4 e_g^2$ **(1/2)**

This complex is paramagnetic due to the presence of four unpaired electrons. **(1/2)**

(ii) Dichloridobis(ethane-1,2-diamine)cobalt(III) nitrate **(1)**

OR

(i) In $[Ni(CN)_4]^{2-}$: Ni is present as Ni(II) with $3d^8$ configuration.



∴ The complex ion has a square planar geometry. **(1)**

(ii) Cu(Z = 29) : [Ar] $3d^{10}4s^1$

Cu^{2+} : $3d^9$

Number of unpaired electron = 1

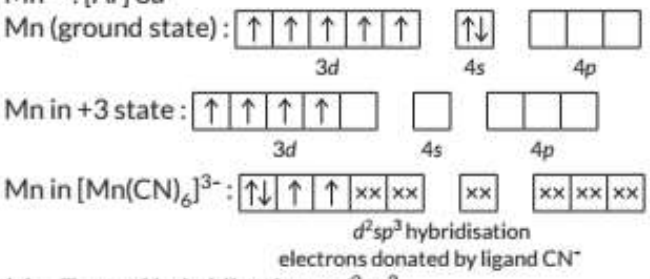
Magnetic moment, $\mu_B = \sqrt{n(n+2)}$

$= \sqrt{1(1+2)} = 1.73$ B.M. **(1)**

7. $[Mn(CN)_6]^{3-}$

Mn : [Ar] $3d^5 4s^2$

Mn^{3+} : [Ar] $3d^4$



(a) Type of hybridisation – d^2sp^3 **(1)**

(b) Magnetic moment value =

$\sqrt{n(n+2)} = \sqrt{(2(2+2))} = 2.83$ B.M.

(n = no. of unpaired electrons) **(1)**

(c) Type of complex – inner orbital complex. **(1)**

8. (a) The colour of coordination compounds depends upon the type of ligands and *d-d* transition taking place. H_2O is weak field ligand, which causes small splitting, leading to the *d-d* transition corresponding green colour, however due to the presence of ethane-1,2-diamine (*en*) which is a strong field ligand, the splitting is increased. Due to the change in t_{2g} – e_g splitting the colouration of the compound changes from green to blue. **(1)**

(b) Formula of the compound is $[Co(H_2NCH_2CH_2NH_2)_3]_2(SO_4)_3$. **(1)**

The hybridisation of the compound is d^2sp^3.

Cobalt in this complex has an oxidation number of +3.

The electronic configuration of Co^{3+} : $3d^6$ $4s^0$ and *en* is a strong field ligand that will cause pairing of electrons in *d*-orbital of Co^{3+}.

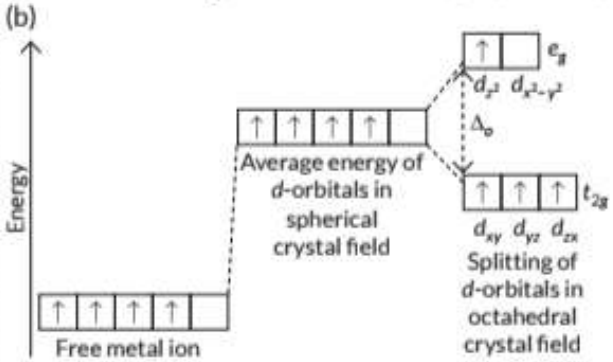


(1)

OR

(a) As the fourth electron enters one of the e_g orbitals giving the configuration $t_{2g}^3 e_g^1$, *i.e.*, no pairing of electrons which indicates $\Delta_o < P$ hence, it is a high spin complex. **(1)**

(b)



(2)